



European M&A Activity: Influences of Macroeconomic Indicators and Covid-19

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Abstract: The purpose of this study is to explore how macroeconomic indicators have affected M&A activity in European markets over a period that includes both the pre-Covid period and the Covid-19 pandemic. By analyzing these factors, the study aims to provide insights into how macroeconomic conditions and unprecedented events like the Covid-19 pandemic influence M&A activity. Previous studies on M&A activity have primarily focused on the impacts of macroeconomic indicators under normal market conditions and during financial crises. However, the Covid-19 period has not yet been studied in this context. This study aims to fill this gap by examining how macroeconomic indicators influenced M&A activity during both typical conditions and the extraordinary circumstances caused by the pandemic. This includes the impact of governmental lockdowns, changes in consumer spending, and shifts in employment structures. The study examines M&A transactions completed between January 2017 and December 2022, focusing on deals where a majority stake was acquired. Data was collected on European companies as acquirers without geographical limitations for the target companies. Multiple linear regression analysis (Ordinary Least Squares) was utilized to test the effects of macroeconomic indicators, including GDP, interest rates, stock market performance, unemployment rates, and exchange rates, on M&A activity by examining the number of transactions, the value of transactions, and the ratio of intra-European transactions to total transactions as dependent variables. A dummy variable for the Covid-19 period was included to capture pandemic-specific effects. The results provide empirical evidence that macroeconomic indicators significantly influence M&A activity in European markets. GDP growth stimulates M&A, while higher interest rates unexpectedly correlate with increased transaction value. Additionally, stock market performance significantly drives M&A, and higher unemployment rates also show positive correlations. Lastly, The Covid-19 pandemic had a negative impact on M&A activities.	
Keywords: M&A, Covid-19, Macroeconomic indicators, European markets, Crisis, Mergers and Acquisitions, GDP, interest rates, stock market, unemployment rates, exchange rates	

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Sammandrag: Syftet med denna studie är att undersöka hur makroekonomiska indikatorer har påverkat M&A-aktivitet på de europeiska marknaderna under en period som inkluderar både tiden före Covid-19 och under pandemin. Genom att analysera dessa faktorer syftar studien till att ge insikter om hur makroekonomiska förhållanden och extraordinära händelser som Covid-19-pandemin påverkar M&A-aktivitet. Tidigare studier om M&A-aktivitet har främst fokuserat på effekterna av makroekonomiska indikatorer under normala marknadsförhållanden och under finanskriser. Dock har Covid-19-perioden ännu inte studerats i detta sammanhang. Denna studie syftar till att fylla detta gap genom att undersöka hur makroekonomiska indikatorer påverkade M&A-aktivitet under både typiska förhållanden och de extraordinära omständigheterna som pandemin medförde. Detta inkluderar effekten av statliga nedstängningar, förändringar i konsumentutgifter och skift i anställningsstrukturer. Studien undersöker M&A-transaktioner som slutfördes mellan januari 2017 och december 2022, med fokus på transaktioner där en majoritetsandel förvärvades. Data samlades in på europeiska företag som förvärvare utan geografiska begränsningar för målbolagen. Multipel regressionsanalys (Ordinary Least Squares) användes för att testa effekterna av makroekonomiska indikatorer, inklusive BNP, räntor, aktiemarknadens utveckling, arbetslöshetsnivåer och valutakurser, på M&A-aktivitet genom att undersöka antalet transaktioner, värdet av transaktioner och andelen intraeuropeiska förvärv i förhållande till totala transaktioner som beroende variabler. En dummyvariabel för Covid-19-perioden inkluderades för att undersöka pandemispecifika effekter. Resultaten ger empiriska bevis för att makroekonomiska indikatorer signifikant påverkar M&A-aktivitet på de europeiska marknaderna. BNP-tillväxt stimulerar M&A, medan högre räntor oväntat korrelerar med ökade transaktionsvärden. Dessutom driver aktiemarknadens utveckling signifikant M&A, och högre arbetslöshetsnivåer visar också positiva korrelationer. Slutligen hade Covid-19-pandemin en negativ inverkan på M&A-aktiviteter.	
Nyckelord: M&A, Covid-19, Makroekonomiska indikatorer, Europeiska marknader, Kris, Fusioner och förvärv, BNP, räntor, aktiemarknad, arbetslöshetsnivåer, valutakurser	

TABLE OF CONTENTS

1	INTRODUCTION	1
1.1	Contribution of the thesis	2
1.2	Purpose of the study	2
1.3	Scope of the study	3
1.4	Structure of the thesis	4
2	MERGERS AND ACQUISITIONS	5
2.1	History and evolution of M&A	5
2.1.1	Merger waves	6
2.2	Terminology	7
2.3	Types of mergers	8
2.4	The Covid-19 and global M&A	8
3	MOTIVES FOR M&A	11
3.1	Strategic- and common objectives	11
3.2	The economic disturbance theory	12
3.3	The capital market conditions	13
3.4	The agency theory	13
3.5	The hubris theory	14
3.6	The overvaluation hypothesis	14
4	LITERATURE REVIEW	16
4.1	Overview of the literature	16
4.2	Beckett (1986)	17
4.2.1	Data and methodology	18
4.2.2	Results	18
4.3	Harford (2005)	19
4.3.1	Data and methodology	19
4.3.2	Results	20
4.4	Resende (2008)	20

4.4.1 Data and methodology	21
4.4.2 Results.....	21
4.5 Choi & Jeon (2011).....	22
4.5.1 Data and methodology	22
4.5.2 Results.....	22
4.6 Uddin & Boateng (2011).....	23
4.6.1 Data and methodology	23
4.6.2 Results.....	24
4.7 Summary of the previous research	24
5 DATA.....	27
5.1 Micro- and macroeconomics	27
5.1.1 Macroeconomic indicators	28
5.2 The data collection and transformation	28
5.2.1 M&A data collection	29
5.2.2 Macroeconomic data collection	31
5.3 Variable description.....	33
5.3.1 Dependent variables.....	33
5.3.2 Independent variables	33
5.3.3 Natural logarithm of the variables.....	34
5.4 Descriptive statistics	35
5.5 Correlation matrix.....	36
6 METHODOLOGY	38
6.1 Hypotheses formulation.....	38
6.1.1 Statistical hypotheses	42
6.2 Multiple regression analysis	43
6.3 Model diagnostics	45
6.3.1 Normal distribution in the error terms.....	46
6.3.2 Multicollinearity	46
6.3.3 Heteroscedasticity.....	47
6.3.4 Autocorrelation.....	47
7 EMPIRICAL RESULTS.....	48
7.1 Macroeconomic indicators' effect on the number of transactions.....	48

7.1.1 Normal independent variables	48
7.1.2 Lagged independent variables	49
7.2 Macroeconomic indicators' effect on the value of transactions.....	50
7.2.1 Normal independent variables	50
7.2.2 Lagged independent variables	51
7.3 Macroeconomic indicators' effect on the ratio of intra-European transactions.....	53
7.3.1 Normal independent variables	53
7.3.2 Lagged independent variables	54
7.4 Effects of Covid-19	55
7.5 Summary of the results	57
8 DISCUSSION	59
8.1 The macroeconomic indicators' effect on M&A	59
8.1.1 Gross domestic product	59
8.1.2 Interest rates	59
8.1.3 The stock market	60
8.1.4 The unemployment rates	61
8.1.5 The exchange rates	62
8.1.6 Non-lagged versus lagged indicators	62
8.1.7 Positive domestic changes yield more domestic activity	63
8.2 Effects of Covid	64
9 CONCLUSION	65
9.1 Limitations of the study	66
9.2 Suggestions for future research	66
9.3 The use of artificial intelligence in the text.....	67
REFERENCES	68
APPENDIX 1	74

TABLES

Table 1 Summary of previous research	25
Table 2 M&A transaction data collection.....	30
Table 3 Macroeconomic indicator data collection	32

Table 4	Descriptive statistics	35
Table 5	Correlation matrix.....	37
Table 6	Summary of research hypotheses	42
Table 7	Results – Number of transactions.....	50
Table 8	Results – Value of transactions.....	53
Table 9	Results – Ratio of European targets (intra-European transactions divided by all transactions)	55
Table 10	Results – The impact of Covid-19.....	56
Table 11	Results – Summary	57
Table 12	Results – Summary of the relationship direction.....	58

FIGURES

Figure 1	Historical illustration of M&A activity	6
Figure 2	The final M&A data sample	31

1 INTRODUCTION

The field of mergers and acquisitions (M&A) is a critical component of the global economic landscape, facilitating corporate growth and market efficiency (Hossain, 2021). Data from Statista (2024) indicates a significant increase in M&A transactions in the recent decades, revealing the cyclical nature of these activities moving in line with economic fluctuations. Global number of M&A deals have risen from 2 676 deals in 1985 to 57 948 deals in 2021, which was the all-time high year (Statista, 2024). This growth in the number of deals demonstrates the importance of M&A and the strategic utilization of M&A as a means for companies to consolidate market positions, diversify their operations, and acquire strategic assets that can enhance their competitive edge in various industries.

In addition, the fluctuation in transaction values and the increase in M&A activity suggest that economic incentives to reallocate assets and pursue market efficiency have increased, despite risks and uncertainties. These changes are likely to reflect fluctuations in macroeconomic conditions such as business cycles, productivity, monetary and fiscal policies, as well as trends in stock- and bond markets and overall demand. This macroeconomic framework shapes microeconomic and operational decisions at the firm level and influences factors such as leverage, cash flow and liquidity, which are crucial for the implementation of M&A strategies. (Choi et al., 2011)

The growth of the M&A market has led to its increased research. According to Malik et al. (2014), most of the studies in the industry focus on understanding the importance and reasons behind entering the market. Also, merger waves and cyclicalities have been studied extensively (Golbe & White, 1993; Rhodes-Kropf & Viswanathan, 2004; Harford, 2005; Resende, 2008). Although the merger waves and cyclicalities might include and reveal that the turning points of the waves are due to economic shifts, there is a lack of research in the literature on how the M&A activity is affected by different crises and further, what role does different macroeconomic variables play in it. Macroeconomic indicators are also widely studied in explaining M&A activity, often with a focus on cross-border transactions. In this context, Ibrahim and Raji (2017) focused on Asian markets, Choi and Jeon (2011) on the U.S. markets, Uddin and Boateng (2011) on the U.K. and Erel et al. (2012) on global M&A, to name a few.

Grave et al. (2012) studied the M&A activity during the global financial crises in a qualitative study with a focus on cross-border transactions, which stated that the M&A market was trending up before the sub-prime crises and shortly after initial signs of the crises, lending slowed and dried up sources of funding for the transactions. Exploring the impact of

macroeconomic factors on M&A activity, especially during the Covid-19 pandemic, is crucial for understanding corporate strategies during economic shocks as Beltratti & Paladino (2013) refer to Cai et al. (2011) research, which highlights the significance of analyzing M&A within different economic environments.

1.1 Contribution of the thesis

The previous literature on mergers and acquisitions has thoroughly examined various determinants of M&A activity but has not fully addressed the unique conditions brought by crisis periods or the recent Covid-19 pandemic. Thus, the influence of macroeconomic indicators on M&A activities during crisis periods remains particularly understudied. Also, while the focus has been on various geographical markets, the European market has not been studied as extensively as the others. While the work of Vasconcellos & Kish (1996) on cross-border M&As and Chiriac (2021) on the impact of economy on European M&As has contributed valuable knowledge, these studies have not covered crises periods, or the Covid-19 period nor focused on market reactions during a crisis of such global extent. It is evident that there is no literature available on this particular problem area, which justifies further investigation.

The pandemic has led to unprecedented economic challenges that have not been captured in previous research. These include the impact of governmental lockdowns, changes in consumer spending, and shifts in employment structures. Having these previously unknown factors in the study could yield insights with implications for future policy development and strategic planning in M&A activities. Thus, this thesis aims to analyze in depth how macroeconomic indicators shape M&A activity during both typical market conditions and the extraordinary conditions caused by the Covid-19 pandemic. By conducting an analysis of the effects of macroeconomic indicators on M&A activity before and during the pandemic, this paper intends to expand the existing knowledge and provide a clearer understanding of the M&A market's behavior in response to significant economic disruptions. Moreover, the study may offer guidance on how macroeconomic factors' influence on M&A operations has evolved due to the pandemic, helping different stakeholders in adapting to a rapidly changing economic environment.

1.2 Purpose of the study

Understanding market dynamics in the context of mergers and acquisitions requires an understanding of the interaction between macroeconomic forces and corporate strategies. Macroeconomic factors have historically influenced M&A activity in European markets in predictable ways that matched company cycles and general economic trends (Chiriac 2021).

However, the Covid-19 has brought previously unknown variables to the economic environment, possibly affecting the traditional patterns of M&A activity.

This study seeks to explore how macroeconomic indicators have affected M&A activity in European markets by examining the number of transactions, the value of transactions, and the ratio of intra-European transactions to all transactions, over a time period that includes both the pre-Covid period and the Covid-19 pandemic. The following research questions are formulated:

- 1. How do European macroeconomic indicators explain M&A activity?*
- 2. How do the European macroeconomic indicators explain the tendency to acquire companies from Europe?*
- 3. Does the Covid-19 pandemic have an impact on the overall M&A activity?*
- 4. Does the Covid-19 pandemic have an impact on the intra-European M&A activity?*

These research questions were developed based on a review of existing literature and theoretical frameworks. Macroeconomic factors such as GDP growth, interest rates, stock market performance, unemployment rates, and exchange rate fluctuations have been shown to significantly influence corporate investment decisions and M&A activity. However, the Covid-19 pandemic introduced new uncertainties and potential disruptions to these established patterns.

The rationale to these questions arises from the necessity to understand both the general and specific impacts of macroeconomic variables on M&A activities. The first and second questions address traditional economic indicators' roles in shaping M&A trends, while the third and fourth questions specifically consider the pandemic's impact, reflecting the challenges and changes it brought to the economic landscape.

1.3 Scope of the study

This thesis examines mergers and acquisitions where a majority stake is acquired, focusing on transactions that have been completed. The geographical scope for acquiring companies is limited to companies located in Europe. There are no geographical limitations for the target companies. The data collection period spans from January 1, 2017, to December 31, 2022, and

includes only deals valued at or above €1 million. The study excludes the financial services and banking industries. The analysis has been conducted utilizing regression analysis to understand the impact of various macroeconomic indicators on M&A activity.

1.4 Structure of the thesis

This thesis continues by examining mergers and acquisitions as a whole in chapter two. Chapter three briefly presents the motives and financial theories related to mergers and acquisitions. Chapter four reviews previous research findings on the topic. Chapter five describes the data, how it was gathered, and the rationale behind it. The methodology section in chapter six explains how the study was executed, including the presentation of the empirical method used. Chapter seven presents the empirical results of the study. Chapter eight discusses these empirical findings, comparing the results to previous research. The thesis concludes with a summary of the main findings, a discussion of the study's limitations, and suggestions for future research.

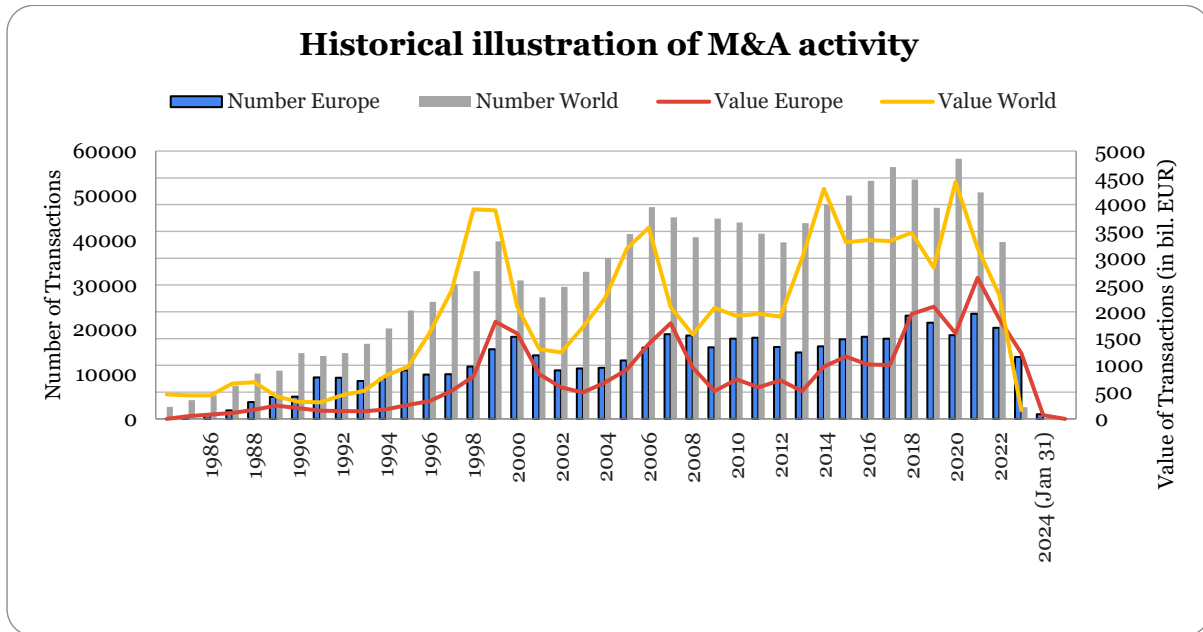
2 MERGERS AND ACQUISITIONS

This chapter reviews the history and evolution of mergers and acquisitions (M&A), examining their development from the 18th century in the United States and the 19th century in Europe. It explores key periods known as merger waves, highlighting significant trends and motivations driving M&A activities during these times. The chapter also discusses the different types of mergers, including horizontal, vertical, and conglomerate mergers, and their strategic objectives such as achieving economies of scale, securing supply chains, and diversifying product lines. Additionally, it examines the distinct phases of M&A activities influenced by economic conditions, technological advancements, and regulatory changes. Lastly, the impact of the Covid-19 pandemic on global M&A is analyzed.

2.1 History and evolution of M&A

The history of mergers and acquisitions started in the 18th century in the United States and in the 19th century in Europe (Focarelli et al., 2002, as cited in Malik et al., 2014). Most research on M&A has been done in the US and Europe, with much less in developing countries. For the last 30 years, companies have used M&A as a key strategy for restructuring. The developed countries saw this trend first, especially the US and UK, but later, developing countries began to use M&A as well. The growth of M&A is shown by the fact that in the US, the number of M&As tripled in the last decade of the 20th century, and their value increased fivefold. (Malik et al., 2014).

Figure 1 presents a historical overview of M&A activity from 1985 to 31 January 2024, comparing Europe to global data. The number of transactions is indicated by bars, with Europe in blue and the world in grey, suggesting a higher frequency of deals on a global scale, as global data also includes Europe. The lines represent the value of transactions in billions of EUR, with Europe in red and the world in yellow, highlighting the monetary size of M&A activities. Peaks and troughs in both the number of transactions and their values are apparent, reflecting the dynamic nature of the M&A market influenced by varying economic conditions and cycles over the years. The chart demonstrates that while the number of deals fluctuates over time, the overall trend is upward, signifying an active M&A market. Similarly, the value of deals, particularly on a global scale, shows significant spikes, indicating periods of especially high-value transactions.

Figure 1 Historical illustration of M&A activity

Source: Institute for Mergers, Acquisitions and Alliances (26 February 2024) Thomson Financial, Capital IQ

2.1.1 Merger waves

The evolution of mergers and acquisitions can be traced through six significant periods, each marked by distinct trends and motivations. The journey began in 1897, stretching to 1904, where M&A activities grew among manufacturing firms. These companies, aiming to dominate markets in sectors such as railroads and power, embarked on horizontal mergers. Despite their ambitions, many of these early consolidations failed to meet their objectives, as highlighted by Fatima & Shehzad (2014).

A shift in focus occurred in the beginning of the second wave from 1916 to 1929, where the aim was to form oligopolies rather than monopolies. This era was characterized by technological advancements in transportation, seeing a blend of horizontal and conglomerate mergers across industries like ore, food, oil, and chemicals. Financial institutions, particularly banks, played a key role by facilitating these deals with easy loan terms. However, this wave ended with the stock market crash in 1929. (Golubov & Petmezas, 2013)

Between 1965 and 1969, a third wave began, mainly involving conglomerate transactions, financed mostly by owners' capital, which reduced dependence on banks. However, mergers of different types of firms started to yield less and less profit by 1968 (Fatima & Shehzad, 2014). The fourth wave, covering the years 1981-1989, was significant because of the increased level of hostile takeovers, which by then had become an accepted expansion strategy. The

nature of these takeovers, whether friendly or hostile, was largely determined by the target company's board's response. This period witnessed key mergers in sectors such as oil and gas, pharmaceuticals, banking, and airlines (Golubov & Petmezas, 2013).

Entering the fifth wave from 1992 to 2000, the M&A landscape was influenced by a global stock market boom and deregulation, leading to significant activities in the banking and telecom sectors. This phase leaned towards financing deals with equity capital over debt (Kouser & Saba, 2011). Lastly, according to Malik et al. (2014), the most recent, sixth wave from 2003 to 2007, saw mergers in a diverse range of sectors including metals, oil and gas, utilities, telecoms, banking, and healthcare. Driven by globalization and supported by governmental policies in countries like France, Italy, and Russia, this wave aimed at creating strong national and international entities. Private equity firms played a significant role, with a notable increase in cash-financed deals, facilitated by the ready availability of credit at low interest rates (Alexandridis, 2012, as cited in Malik et al., 2014). Overall, this chronological order of presenting the merger waves highlights the dynamic nature of M&A, shaped by economic conditions, technological developments and regulatory changes that reflect companies' evolving growth strategies and market power.

The most recent merger wave to be empirically documented is the sixth, spanning from 2003 to 2007. According to my research, there are no studies yet on the 7th merger wave. This may be because it is still in progress or has not been empirically confirmed. However, merger data suggest that in 2013, the number of mergers began to increase significantly, both in terms of number and transaction value. This growth peaked in 2014, followed by a slight decline from 2015 to 2020. In 2020-2021, merger activity surged again, reaching an all-time high before experiencing a sharp decline. The M&A data suggests that the seventh merger wave likely took place between 2013 and 2021, peaking in 2021 with over 58,000 transactions valued at more than €4.4 trillion worldwide. This hypothesis is expected to be confirmed by future academic research in the near future.

2.2 Terminology

While mergers and acquisitions are distinct concepts, they're often used as if they're the same. An acquisition involves one company buying part or all of another company. In contrast, a merger occurs when two or more companies join forces to form a new, single entity. This process is a legal action where only one of the companies emerges as a surviving legal entity. Essentially, mergers unite firms into a single company, whereas acquisitions typically involve a larger, financially stable company buying out a smaller one. These activities, including

takeovers and corporate restructuring, fundamentally alter the ownership and control structures of the companies involved. (Malik et al., 2014)

2.3 Types of mergers

Berk et al. (2015) classify mergers based on their characteristics and objectives. One common type is horizontal mergers, which occur between companies in the same industry. These mergers often aim to achieve economies of scale, increase market share, or reduce competition. Another type is vertical mergers, which involve firms operating at different stages of the production process within the same industry. This type of merger allows a company to secure its supply chain or distribution network, reducing costs and increasing efficiency. Conglomerate mergers are mergers that take place between companies engaged in unrelated businesses. Such mergers may be motivated by diversification of product lines, markets or investment portfolios, which may reduce the overall risk of the business.

In addition, a distinction can also be made between mergers involving market expansion and product expansion. Market extension mergers are between firms that offer the same products but operate in different markets. Their aim is to gain access to a wider market base. In contrast, product-enhancing mergers are between companies that sell different but related products in the same market and aim to combine product offerings and services to existing customers. (Berk et al., 2015)

The complexity and structure of mergers can vary significantly, with some involving straightforward stock-for-stock transactions, while others may use a combination of stock, cash, or other instruments. In some cases, the terms of the merger may include special provisions, such as earnouts, which are contingent payments based on the future performance of the target company, or stock swaps, in which shareholders of one company receive shares in the merging company. (Berk et al., 2015)

2.4 The Covid-19 and global M&A

The Covid-19 pandemic started in late 2019 in the city of Wuhan, China. The World Health Organization (WHO) declared the disease-causing SARS-CoV-2 virus public health emergency of international concern (PHEIC), WHO's highest level of alarm, on 30 January 2020, and it was later characterized as a pandemic on 11 March 2020 (WHO, 2024). The rapid spread and severity of the disease led countries around the world to implement varying degrees of lockdowns and restrictions to control the spread of the virus.

Kooli and Son (2021) describe the global impact of Covid-19 to global M&A in their research paper *Impact of COVID-19 on Mergers, Acquisitions & Corporate Restructurings*. Their study collects statistical data on global transaction volume, value, and type to understand how mergers, acquisitions, and other restructuring activities have been leveraged to achieve corporate objectives during the unforeseen time of a global pandemic.

They state that the announcement of Covid-19 caused a 50% drop in the number of global acquisitions by mid-February, due to the unpredictable business environment caused by backlogs of orders and uncertainty related to the pandemic. Despite these challenges, the total value of announced acquisitions from January to October 2020 was \$2.2 trillion. The recovery was supported by a slow easing of lockups and optimism generated by the vaccine development, which helped to re-launch the M&A sector (Kooli & Son, 2021). They continue that, the second half of 2020 saw a significant recovery, with the fourth quarter being the third strongest for M&A in two decades, marked by \$612 billion of deals, an increase from \$461 billion in the same period in 2019, according to the Financial Times.

The total number of transactions in 2020 fell by around 7.5% from the previous year, and the total value of transactions fell by 16%. However, according to PwC on the Global M&A Industry trends, the second half of the year saw a significant increase in both number and value, up 94% and 18% respectively, surpassing the second half of 2019. This increase was primarily driven by an increase in the number of mega-deals (> \$5 billion), particularly in the technology and telecoms sectors, where deal volumes increased by 118% and 300% respectively, highlighting the sector's ability to take advantage of favorable interest rates and scaling, which can be more effective during economic downturns. (Kooli & Son, 2021)

In terms of divestitures, during the first 12 months of the pandemic, many companies strengthened their capacity by increasing debt (Kengelbach et al., 2020). However, as markets started to stabilize, there was a significant shift towards divestments to improve the financial situation and prepare for strategic growth. According to a survey by the M&A Leadership Council, 23% of respondents considered divesting for these reasons. This trend is expected to continue as companies adjust to the post-pandemic reality and emphasize strategic restructuring and financial sustainability. (Kooli & Son, 2021)

The authors conclude that there is cautious optimism in the M&A market, driven by changes in corporate priorities because of societal changes. Given the availability of debt and equity financing, the market is expected to remain strong. It will also be driven by baby boomers selling their businesses to avoid potential increases in capital gains taxes, and the need to

integrate technology across all business sectors. Despite the flexibility shown, the ongoing Covid-19 pandemic requires M&A stakeholders to strengthen their defensive strategies against such external threats and to prioritize technological developments. While this study takes into account some new trends, it recognizes that it does not fully capture the multiple impacts of the pandemic on global M&A activity but gives a relatively good view on the sudden changes in the market. (Kooli & Son, 2021)

3 MOTIVES FOR M&A

For smaller organizations or those with lower profits, M&A often serve as a vital strategy to remain competitive and survive in emerging markets. These global business strategies allow companies to penetrate potentially lucrative new markets or diversify into new business areas (Malik et al., 2014). As Georgios & Georgios (2011) point out, the main motivation for engaging in M&A is to exploit synergies that go beyond the potential of operating independently. M&A have been shown to increase shareholder wealth and reduce operating costs, highlighting their importance in optimizing financial performance. Moreover, Gattoufi et al. (2009) argue that management favors M&A because these activities expand their powers and provide opportunities to achieve both immediate and long-term corporate goals. In this chapter, the reasons behind engaging in the M&A process are divided into two subheadings: strategic objectives and common objectives, which provide detailed insights into the most common motivations found in the literature. Additionally, individual theories influencing M&A activity are presented.

3.1 Strategic- and common objectives

The first topic that appears is synergies. They arise from the merger of two companies, generating a value that surpasses what each could have achieved independently. This is the additional worth that comes into being when the companies combine their strengths and capabilities. Vertical integration occurs when a company expands its control over multiple stages of production or within its supply chain. This can involve merging with suppliers or distributors to create a more streamlined and cost-effective process from production to consumer delivery. (Berk et al., 2015)

Access to specialized knowledge is another key driver for M&A. Companies may look to acquire or merge with firms that have the expertise they lack. This can be a strategic move that strengthens a company's competitive advantage by integrating specific skills and knowledge. The advantages of diversifying are often pointed out as the rationale behind conglomerate mergers. Such benefits include expanded borrowing capacity and improved cash flow, as well as more effective risk management. Consequently, these factors may contribute to a reduction in the company's expenses. (Berk et al., 2015)

Economies of scale and -scope are among the first “common objectives” behind the M&A deals rationale. Economies of scale are the cost-saving advantages that emerge when a company increases its production volume. These advantages, which typically reduce the cost per unit, are particularly pronounced for larger companies, which can produce goods more efficiently

than their smaller counterparts. Economies of scale, in turn, are achieved when a company combines the marketing and distribution of several related products, thereby reducing costs. By offering a diversified range of products, a company can spread its fixed costs over a wider base. (Berk et al., 2015)

Additionally, Berk et al. (2015) describes three other objectives, which relate to possible monopoly positions of companies, tax refunds and managerial motives to M&A. Mergers and acquisitions can sometimes lead to less competition in an industry, which can result in a monopoly position. This is a double-edged sword, as the results can be advantageous for the company concerned, increasing its market power. However, these practices are often regulated by competition laws, which are designed to maintain fair competition and prevent the formation of monopolies. When a company makes a profit, it is liable to pay taxes on it. However, if the company suffers losses, it may be entitled to a tax refund. This can lead to a situation where losses in one part of the company can mitigate tax liabilities arising from profits in another segment. Lastly, considering managerial motives to merge shows that although economic factors are commonly cited as a justification for mergers, there are cases where the personal interests of the manager may play a role. This may be the case, particularly when the manager acts after observing a decline in the share price of the publicly traded target company. Such actions may be motivated by personal prestige and influence or by an overly optimistic belief in the ability to improve the performance of the target company.

3.2 The economic disturbance theory

The economic disturbance theory, proposed by Gort (1969), suggests that macroeconomic disturbances can alter business valuations by changing future cash flow expectations, potentially leading to valuation differences between shareholders and investors. Furthermore, the theory suggests that these valuation changes do not necessarily arise from increased investor uncertainty, but rather from the decreased reliability of historical data in predicting future outcomes. As the common knowledge base among investors weakens, a wider range of future predictions emerges. Therefore, if an economic disturbance aligns the average valuation with that perceived by owners or managers, the increased range of valuations could lead to some prospective buyers to assign higher values to a company's assets than those currently held by its controllers. Typical triggers for such economic shocks include sudden technological developments and fluctuations in securities prices. According to this theory, such valuation differences, particularly when investors assign higher valuations than shareholders post-disturbance, may trigger M&A activities. This relationship is also supported by broader economic indicators such as GNP and GDP, which illustrate how macroeconomic shifts influence M&A trends (Resende, 2008)

3.3 The capital market conditions

The theories developed by William Sharpe and Eugene Fama, although not directly focused on M&A, are essential for understanding the financial aspects and market behaviors related to the transactions in the M&A market. The Capital Asset Pricing Model (CAPM), introduced by Sharpe (1964), provides a way to calculate the expected return on an investment by considering its risk in relation to the market. According to CAPM, the expected return on a security is equal to the risk-free rate added to the product of the security's beta and the anticipated market risk premium. This model lays the ground for security valuation and is important for M&A analysis as it helps determine the risk-adjusted returns of acquiring companies, providing a financial baseline to assess whether the acquisition should generate sufficient returns to justify its risks.

According to Fama's (1970) Efficient Market Hypothesis (EMH), stock prices reflect all the information available at a given point in time, which makes the market "informationally efficient". In context of M&A, this hypothesis is applied to understand how quickly the market prices in the information regarding potential or actual mergers, reflecting these in the stock prices of the companies involved. EMH helps analyze the immediate impact of M&A announcements on company valuations and assesses whether the transactions create shareholder value, as any potential benefits should be instantly obtained in the stock prices. Both theories presented are interconnected in the context of M&A, where CAPM helps evaluate the cost and benefits of M&A based on risk assessments and financing conditions, while EMH provides a way to see how these actions are reflected in market prices.

3.4 The agency theory

According to Eisenhardt (1989), the agent theory focuses on principal-agent relationships to organize information efficiently and deal with the costs of risk bearing. It analyses the contracts between a principal, such as a shareholder, and an agent, such as a manager. It is based on the assumptions that individuals, acting in their self-interest, are rationally limited and avoid risk. The theory recognizes a potential goal conflict between the parties in an organization and considers the managing of information asymmetry between principal and agent as a measure of efficiency. Agent theory treats information as a commodity that can be purchased and addresses problems such as moral hazard and adverse selection associated with risk sharing. The theory is particularly relevant in situations where the principal and agent have different objectives and risk preferences, which affects organizational governance such as hiring, regulation, and general management practices. (Eisenhardt, 1989)

In M&A transactions, these conflicts can occur when managers seek acquisitions that enhance their personal position or control, rather than those that maximize shareholder value. For example, managers may favor acquisitions that increase the size of the company, as their compensation is often tied to measures such as revenue or assets under management, rather than profitability or long-term value creation. According to agent theory, such conflicts can lead to inefficient M&A decisions unless there are mechanisms in place to align managers' interests with those of shareholders.

3.5 The hubris theory

The hubris hypothesis, referring to Roll (1986), proposes that decision-makers in acquiring firms often overvalue their targets, leading them to pay too much. This hypothesis argues that this overpayment is driven by the acquirer's overconfidence (hubris), rather than by the underlying value of the target company. According to this theory, the acquiring firm's management believes so strongly in their own capability to enhance the target's value that they ignore indications that the bid price exceeds the actual market value of the target. (Roll, 1986)

This hypothesis further suggests that when the acquirer's valuation of a target company falls below the market price, the bid is usually abandoned, reflecting a rational choice. However, when the valuation exceeds the market price, bids often proceed, driven by the acquiring management's confidence in their own assessment and capabilities. This behavior will continue despite the potential for negative financial impacts, suggesting that the decision-making process in these cases may be influenced more by the acquirers' confidence than by rational investment evaluation. The hubris hypothesis serves as a null hypothesis for corporate takeovers, in which all market participants are strongly efficient, meaning they fully incorporate and act on all available information. (Roll, 1986)

3.6 The overvaluation hypothesis

The overvaluation hypothesis, according to Rhodes-Kropf and Viswanathan (2004) suggests that stock market valuations significantly influence M&A, particularly the choice of shares or cash as the transaction instrument. They argue that merger waves often occur not only due to economic factors like deregulation or technological innovations but also due to market mispricing. In overvalued markets, M&A are more likely as targets may overestimate the value of stock offers, influenced by the inflated market conditions and the value of the acquiring company. Consequently, transactions in such markets frequently involve stock as the medium of exchange, whereas cash is more common in undervalued markets where the valuation is found more accurate. Thus, market mispricing plays a critical role, affecting both the

completion and the form of payment used in M&A transactions. (Rhodes-Kropf & Viswanathan, 2004)

4 LITERATURE REVIEW

This chapter intends to provide an overview of the previous literature in the field of mergers and acquisitions focusing on macroeconomy and how it affects the activity in the M&A market. The overview is followed by a more detailed presentation of five individual studies including their data and methodologies and results, as well as a table that summarizes the presented studies.

4.1 Overview of the literature

The academic literature has covered the topic extensively, providing a thorough understanding of the macroeconomic factors that influence mergers and acquisitions. Observing the literature, the focus has primarily been on three factors: the characteristics of individual firms, particular industries, and broader economic or financial market conditions. Key subject to note, which Harford (2005) findings also suggest, is that M&A activity is not randomly distributed but concentrated within industries, proposing that macroeconomic state influences companies' investment behavior.

Historically, interest rates have played a central role in these analyses as a linking variable to M&A activity. Steiner (1975) was one of the first researchers to emphasize the impact of interest rates on M&A, and Guerard (1989) has later supported this observation. Becketti's (1986) analysis also recognized the importance of real interest rates, showing that they explained about one-third of changes in merger activity. More recent analyses by Uddin & Boateng (2011) and Choi & Jeon (2011) further confirm the role of interest rates in shaping M&A trends.

The impact of industrial production, GDP and GNP on merger activity has been extensively studied. Steiner (1975) and Becketti (1986) were pioneers in linking these economic indicators to mergers and acquisitions. Resende (2008) further investigated the relationship between GDP growth and M&A activity and found that economic growth, liquidity, and market prospects are significant drivers of M&A waves. The relationship between these macroeconomic variables and M&A activity was found to be asymmetric and the effects vary across economic situations. Uddin & Boateng (2011) found that GDP growth is correlated with an increase in country CBM&A inflows whereas it has a significant negative impact on outflows.

Stock market performance has been found to be another key factor influencing M&A activity changes. Gort (1969) and Guerard (1989) both found a positive correlation between stock

market conditions and the occurrence of M&A waves, and this result is also supported by studies by Shleifer and Vishny (2003) and Ang and Cheng (2006). However, Harford (2005) questioned this consensus in his study where he approached the topic from the angle of behavioral theory, yet he did not observe a significant link between stock market prices and the frequency of mergers, indicating that the relationship between stock market dynamics and M&A waves might not be as straightforward as previously thought. More recently, Uddin & Boateng (2011) found that domestic stock prices have a positive impact on the outflow and inflows of domestic cross-border M&A (CBM&A) and that a strong economy tends to increase inward investment.

The impact of exchange rates on M&A activity has also been studied. Uddin & Boateng (2011) analyze the impact of exchange rates, along with other indicators such as stock prices and interest rates, on cross-border M&A activity. Their findings show that they have a significant impact on both inward and outward M&A activity. Erel et al. (2012) found that differences in international valuations are often the reason for cross-border mergers and acquisitions, especially when the domestic currency of a firm strengthens, making foreign firms more financially attractive and more accessible. This view is supported by Vasconcellos and Kish (1996), who emphasize the important role of currency strength as a factor influencing M&A decisions.

4.2 Becketti (1986)

Corporate mergers and the business cycle

The study conducted by Becketti (1986) focuses on analyzing the connection between merger activity and the broader business cycle over a span of 25 years, from 1960 to 1985. It investigates how economic expansions and recessions influence the frequency and scale of corporate mergers and how, in turn, these mergers might affect macroeconomic variables like interest rates, money supply, stock prices, and the gross national product (GNP). The researcher uses two models to estimate these relationships, with an emphasis on differentiating the short-term and long-term impacts. They also consider whether mergers could disrupt the normal functioning of credit markets. The conclusion of this study is that merger activity follows the cyclical nature of the economy, increasing in times of economic growth and decreasing during downturns. Furthermore, the study shows that while macroeconomic variables can explain part of the variation in merger activity, the models do not explain a significant part of it, indicating that mergers do not have a measurable feedback effect on the macroeconomic indicators analyzed.

4.2.1 Data and methodology

The study uses two specific measures of merger activity: the number of mergers during a quarter and the total value of assets acquired during the same period. The data for these indicators are from different periods and different sources. For the period 1960-1979, the Federal Trade Commission's Large Merger Series has been used as a data source. Data for later years are from a database compiled by mergers and acquisitions.

The study acknowledges that while the total value of mergers is theoretically a more comprehensive measure, there are concerns about the reliability of this data. Therefore, the analysis uses the number of reported mergers for its greater accuracy. To ensure that the impact of mergers on macroeconomic variables is isolated, the study includes lagged values of these variables in the model. This approach is designed to filter out the simultaneous effect of economic activities that may affect both mergers and macroeconomic and financial variables.

For macroeconomic factors, the analysis includes an index of six key variables: the Standard & Poor's index of stock prices, yield on 3-month Treasury bills, money supply, domestic non-financial debt, capacity utilization rate and gross national product (GNP). These are selected to provide a comprehensive picture of macroeconomic conditions and their potential impact on merger activity. The study distinguishes between short-term and long-term relationships and suggests that short-term relationships may indicate temporary deviations, while long-term relationships may indicate permanent, persistent changes in the variable of interest.

The study formulates a multiple regression model to understand how macroeconomic factors such as interest rates, money supply, stock prices and gross domestic product (GDP) affect the number and value of mergers. It also examines whether merger activity could in turn affect these macroeconomic variables. The model includes lagged values of macroeconomic variables to account for their impact on mergers, thus isolating the effect of mergers from other economic activity.

4.2.2 Results

The study found that about a third of the changes in the total number of mergers can be explained by various macroeconomic conditions. Among the various economic factors, the variation in real interest rates appeared to play the most important role in influencing merger activity. This trend may be due to the dependence of acquiring firms on borrowing. The study also found that in the short run, the extent to which the industry uses its capacity may affect

the number of mergers, suggesting that firms may prefer mergers to investing in new facilities if there is a significant delay in obtaining new equipment.

Contrary to what some expected, the results of the study show that the number of mergers does not affect the economy. Mergers do not appear to have a consistent effect on interest rates or total debt in the economy, although it was previously thought that a link might exist. Thus, concerns that mergers could have negative effects on the broader economy appear to be unfounded.

4.3 Harford (2005)

What drives merger waves?

A study by Harford (2005) examines why industries are subject to merger and acquisition waves. It compares two theories: one in which industry changes lead to mergers due to restructuring (neoclassical model) and another in which managers use misperceptions of the market to acquire assets cheaply (behavioral model). The results of the study support the neoclassical model and emphasize the importance of available capital in driving these waves. The study finds that industry mergers are concentrated over time because of this financial liquidity rather than because of market timing efforts by corporate managers. Capital availability, rather than strategic market timing, is key to understanding why merger waves occur.

4.3.1 Data and methodology

The data used in the study is taken from Thomson Financial's Securities Data Company (SDC) database and includes information on mergers and tender-offer bids worth more than \$50 million between 1981 and 2000. The data are classified by industry, and the study divides the samples into two decades to reflect the large merger waves that were separated by the recession of the early 1990s.

To identify merger waves, the study calculates the 24-month period by industry in each decade during which merger activity was most frequent, and then simulates 1000 random distributions to determine whether actual activity exceeds the 95th percentile point of these simulations. If so, the period is considered a wave. The study found 35 waves in 28 industries, and on average, industries had significantly more bids during wave periods than during non-wave periods, as during wave periods, industries averaged 34.3 bids every 24 months, which is significantly higher than the 7.8 average bids during non-wave periods.

The study uses sequential logit models to predict the start of an industry merger wave using, among other economic variables, the median market-to-book ratio and industry-specific economic shock measures that predict waves only when capital liquidity is high. The study also examines deregulation events and capital liquidity measures to increase the predictive power of the model.

4.3.2 Results

According to the neoclassical model presented in the study, merger waves result from industry reactions to shocks that lead to restructuring through mergers and acquisitions, which causes clustering of such activities. According to the behavioral model, rational managers exploit market mispricing and acquire real assets with overvalued shares. The study directly compares these models and finds support for the neoclassical explanation, especially when modified to consider the role of capital liquidity. The study finds that it is this liquidity that synchronizes individual industry-level merger waves into broader, market-wide waves. The study argues that the link between the valuation level and merger activity, which is central to the behavioral hypothesis, reflects capital liquidity factors rather than market mispricing.

In addition, the study finds that industry merger waves are caused by disruptions, whether economic, regulatory, or technological, and that these waves are not simply the result of business leaders trying to time the market. Adequate capital liquidity is essential to facilitate transactions following these disruptions. Market timing can occur but is not considered to be the driving force behind merger waves. Instead, the cumulative effect of industry-specific mergers triggered by shocks, supported by macro-level capital liquidity, is considered to be the main cause of widespread merger waves.

4.4 Resende (2008)

Mergers and acquisitions waves in the UK: a Markov-switching approach

The study examines the patterns of mergers and acquisitions in the UK from the first quarter of 1969 to the first quarter of 2004, using Markov-Switching models to account for changing circumstances over time. In contrast to traditional models where probabilities remain constant, this study applies models where probabilities can change following the approach of Diebold et al. (1994). The study takes into account economic variables such as growth in GDP, growth in money supply and stock market returns and analyzes their impact on M&A activity. The results suggest that these economic factors affect the changing patterns of M&A waves, and the impact varies across economic conditions. The study finds that a model with variable

probabilities is a better fit than a model with constant transition probabilities, confirming the importance of general economic variables in driving M&A waves, although the effects are asymmetric across different economic conditions.

4.4.1 Data and methodology

The study focuses on the UK and uses quarterly aggregate data to examine mergers and acquisitions from the first quarter of 1969 to the first quarter of 2004. The M&A data, representing the number of domestic companies acquired in the UK, were collected from the Office of National Statistics. Gross domestic product and inflation figures were obtained from the International Financial Statistics of the International Monetary Fund (IFS-IMF). The M4 money supply measure and the FTSE-all shares index, based on Financial Times publications, were also included. From these raw data, three conditioning variables were developed, focusing on the rates of change of these variables, thus avoiding the need to adjust the specific base dates of the series.

The study uses a Markov-switching model with transition probabilities varying over time to examine the impact of economic variables on the pattern of UK M&A activity. This approach allows to examine how these variables may influence the creation of merger waves. Three key variables are constructed and analyzed: GDP, money supply growth and stock market returns. According to the author, the choice of these variables is based on the hypothesis that economic growth can create a favorable environment for mergers, liquidity can influence M&A decisions by affecting the relative cost of financing, and high stock market returns can indicate a favorable long-term business outlook, potentially encouraging M&A activity.

4.4.2 Results

The empirical analysis of the study shows that the M&A environment is ambiguous, characterized by two states of M&A activity, high and normal, and the probability that the economy will remain in its prevailing state. By introducing time-varying transition probabilities into the model, the study extends previous research to examine how economic factors such as GDP growth, money supply growth and stock market returns affect M&A waves. Although significant coefficients on these economic indicators confirm their impact on M&A patterns, their effects are found to be asymmetric. Positive effects are seen in periods of high M&A activity, while periods of low activity have unexpected negative effects. These findings are consistent with previous research showing that M&A activity is persistent and consistent across both states, which is in line with the results of models with constant transition probabilities. Tests comparing the two models suggest that the standard probability model should be rejected in favor of a model that accounts for economic variables as drivers

of M&A waves, especially during periods of lower activity. While this conclusion supports the impact of economic conditions on M&A dynamics, it suggests that the asymmetric effects observed across M&A regimes need to be further investigated.

4.5 Choi & Jeon (2011)

The impact of the macroeconomic environment on merger activity: Evidence from US time-series data.

The study examines the relationship between macroeconomic variables and merger activity in the U.S. between 1980 and 2004. Using data on firm-level mergers and acquisitions and time-series econometric models, a long-run equilibrium between macroeconomic conditions and measures of merger activity is identified. The results show that real income, stock market conditions and monetary policy have a significant impact on the number of M&A transactions, and that the rising phases of business cycles provide optimal conditions for an increase in M&A activity. The post-1998 analysis supports the free cash flow hypothesis and highlights the crucial role of net cash flow of firms as a determinant of merger activity. The study also details short-term adjustment processes towards long-term equilibrium, with significant impulse-lag dynamics ending within 5-6 quarters. In conclusion, the paper highlights the importance of macroeconomic factors in shaping merger activity.

4.5.1 Data and methodology

The methodology used in this study involves an analysis of firm-level M&A data from Securities Data Company's U.S. M&A database. The study period covers mergers announced and completed between 1980 and 2004, focusing on U.S. public and private companies and subsidiaries, when the transaction value was \$1 million or more and when the acquirers ultimately owned 100% of the target's shares. The analysis assessed the overall merger activity using four separate indicators: the frequency of M&A activity, the value of M&A transactions, the value of transactions relative to the trading volume of the S&P 500, and the value of transactions relative to the value of total assets. Macroeconomic variables are GDP, net corporate cash flow, the effective federal funds rate and the S&P 500 index. These were selected based on their potential impact on the financial and operational capacity of firms to undertake M&A transactions. Vector Autoregression (VAR) models were used to examine relationships and identify long-term equilibria and short-term trends.

4.5.2 Results

The results show a strong long-term relationship between macroeconomic indicators and merger activity, indicating that these indicators significantly influence the number and volume

of mergers in the U.S. economy. Specifically, real income affects the frequency mergers occur, while stock market conditions and monetary policy mainly influence the transaction values of these mergers. The study also finds that after economic shocks, merger activity tends to return to this long-term trend within five to six quarters, with the most significant reactions occurring right after the shocks.

Additionally, the analysis points out that different macroeconomic factors have varied impacts during different times of the study, especially during major economic events like recessions or booms. For example, after 1998, the importance of firms' net cash flow increases, supporting the idea that firms with more available cash are more likely to engage in mergers.

4.6 Uddin & Boateng (2011)

Explaining the trends in the UK cross-border mergers & acquisitions: An analysis of macro-economic factors

This study examines the impact of macroeconomic factors on cross-border mergers and acquisitions in the UK between 1987 and 2006. It analyzes how different macroeconomic indicators affect the level of both outward and inward CBM&A activity. Building on previous studies that emphasized industry- and firm-level factors, this study aims to extend the analysis to macroeconomic factors, providing a broader understanding of the determinants of CBM&A trends. The study finds that several macroeconomic factors, such as GDP, interest rates, exchange rates, money supply and stock prices, play a significant role in shaping these activities.

4.6.1 Data and methodology

The study uses data from the Thomson One Banker database and Acquisitions Monthly on UK cross-border M&A transactions between 1987 and 2006. The macroeconomic data is retrieved from the Economist Intelligence Unit and UK National Statistics. The variables included in the analysis are UK GDP, UK money market interest rate, exchange rate of GBP and USD, total value of UK export and import, the FTSE all-share price index, money supply and UK consumer price index. A multivariate regression analysis is used to examine the effects of these macroeconomic variables on inward and outward CBM&A. Moreover, to perform a partial elasticity analysis, the logarithms of the variables are taken, which allows to assess the reaction of the dependent variable to changes in one independent variable while keeping other independent variables fixed. The study also applies Engle-Granger cointegration test and error correction model to deal with stationarity of the data and examine long-run relationships.

4.6.2 Results

The authors show that firms engaged in cross-border transactions can reduce their costs and improve their competitiveness by monitoring the relative levels of stock prices, exchange rates, interest rates and other economic factors that are affected by both home and host country policies. The results suggest that macroeconomic variables significantly affect the inflows and outflows of CBM&A in the UK. Domestic interest rates, exchange rates and stock prices have a positive impact on the outflow of UK CBM&A, while GDP has a significant negative impact. UK CBM&A inflows are primarily influenced by GDP, money supply and stock prices, meaning that the UK's strong economy tends to increase inward investment. The authors state that these findings highlight the important role of macroeconomic variables as determinants of location-specific preferences that influence M&A flows. Moreover, the analysis confirms the long-run equilibrium relationship between these macroeconomic factors and CBM&A activity, which highlights the influence of economic conditions on cross-border investment strategies. Therefore, they conclude that it is crucial that governments carefully manage their macroeconomic policies to reduce the production and transaction costs of multinational firms.

The authors conclude that leveraging macroeconomic factors can provide companies with a competitive advantage. Entering the market before competitors can allow a firm to create uniqueness in the market or use different pricing strategies, which could lead to securing a favorable market position. This suggests that business managers should time their acquisition decisions based on macroeconomic trends and firm- and industry-specific variables to outperform their competitors and increase shareholder value. The models in this study show that accurate awareness of macroeconomic changes can help companies secure a leading position in the market and avoid the pitfalls of simply following industry leaders.

4.7 Summary of the previous research

Previous research on mergers and acquisitions has examined the impact of macroeconomic factors extensively and found an interaction between different macroeconomic indicators and trends in M&A. Interest rates and GDP have consistently been found to be significant predictors of M&A activity, highlighting the role of macroeconomic factors in driving industry M&A trends. Although the correlation between stock market performance and M&A activity is widely recognized, the causality and impact of this relationship remains somewhat unclear. The impact of exchange rates has also played a key role and has been found to have a significant impact on M&A activity. In addition, studies that have looked at variables other than traditional economic indicators have produced mixed results on their role in explaining the M&A activity.

Only a few studies have specifically analyzed the macroeconomic indicators of M&A transactions during times of crisis. This study aims to bridge this gap by providing a deeper understanding of the effects of these indicators both during and outside the Covid-19 pandemic.

Table 1 Summary of previous research

Summary of the previous research				
Authors	Purpose	Time period	Data and Method	Results
Beckett (1986)	Analyze merger activity in relation to business cycles	1960-1985	Merger data from FTC's Large Merger Series (1960-1979) and M&A database; Multiple regression model with macroeconomic variables: index of stock prices, yield on 3-month Treasury bills, money supply, domestic non-financial debt, capacity utilization rate and gross national product (GNP).	Around 1/3 of merger changes are explained by macroeconomic indicators. Real interest rates are crucial. Short-term industry capacity affects the number of mergers. Mergers don't significantly impact macroeconomic variables like interest rates, total debt.
Harford (2005)	Examine drivers behind industry M&A waves	1981-2000	Thomson Financial's Securities Data Company (SDC) data on M&A over \$50M; Industry classification; Differentiate samples by decades; Identify waves with 24-month peak activity & simulation exceeding 95th percentile; Successive logit models, median market-to-book ratios, industry shocks, deregulation, and capital liquidity measures are used	Supports neoclassical model; Capital liquidity drives waves, which turns industry-level merger waves into market-wide phenomena, not market timing. 35 waves in 28 industries. Waves are caused by economic, regulatory, or technological disruptions, not just market timing.
Resende (2008)	Analyze the patterns of M&A in the UK focusing on macroeconomic indicators	1969-2004	M&A data from UK Office of National Statistics; GDP, inflation data from IFS-IMF; M4, FTSE-all share index from Financial Times; Markov-Switching models with time-varying transition probabilities; Analysis of real GDP growth, money supply growth, stock market returns	M&A characterized by two states: high and normal activity; Model with time-varying probabilities provides better fit; Economic growth, liquidity, and market outlook as drivers of M&A waves; Asymmetric impacts of economic factors on M&A, with positive effects in high-activity periods and negative in low-activity; Calls for further study on asymmetric effects.
Choi & Jeon (2011)	Study the impact of macroeconomic indicators on U.S. M&A activity	1980-2004	U.S. M&A data from Securities Data Company; Inclusion criteria: transactions over \$1M and acquirer's 100% ownership; Used VAR models to analyze number of M&As, transaction values, and ratios to S&P 500 trading volume and total assets; Macroeconomic variables include real GDP, corporate cash flow, federal funds rate, and S&P 500 index	Identified a significant long-term correlation between macroeconomic indicators and M&A activity, with real income, stock market conditions, and monetary policy impacting M&A transaction numbers and values; post-economic shocks, M&A activity reverts to long-term trend within 5-6 quarters; Emphasized the increasing significance of firms' net cash flow post-1998 and the differential impact of macroeconomic factors throughout various economic conditions

Uddin & Boateng (2011)	Aims to extend the analysis of how different macroeconomic indicators affect the level of both outward and inward cross-border M&A (CBM&A) activity.	1987 - 2006	Thomson One Banker database and Acquisitions Monthly data for UK cross-border M&A activity. Macroeconomic data from the Economist Intelligence Unit and UK National Statistics. Real GDP, interest rates, exchange rates, trade values, stock index, money supply, and consumer price index are analyzed using multivariate regression to determine their impact on M&A transactions. Logarithmic transformation of these variables helps in conducting partial elasticity analysis to understand their proportional effects. Additionally, Engle-Granger cointegration tests and error correction models are utilized to explore long-term relationships.	CBM&A are significantly influenced by macroeconomic factors such as stock prices, exchange rates, interest rates, and GDP. These variables affect both inflows and outflows, with certain factors, such as interest rates, having a positive effect on outflows, while GDP growth is correlated with an increase in inflows. The study highlights the role of these economic conditions in shaping cross-border investment strategies and suggests that firms can gain competitive advantage by strategically timing acquisitions in response to these macroeconomic trends.
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5 DATA

This chapter first introduces macro- and microeconomics and emphasizes their importance in understanding the general economic environment and economic behavior. The key macroeconomic indicators that are central to analyzing the dynamics of M&A, are discussed. The data collection and transformation processes applied to the data to prepare it for the empirical analysis are presented. The latter sections describe the variables used in the study dividing them into dependent and independent variables as well as presenting a dummy variable. The chapter ends with descriptive statistics of the used data sample and correlation matrix of the variables.

5.1 Micro- and macroeconomics

Macroeconomics and microeconomics are two main areas of economics that deal with different scales of economic activity but are nevertheless interrelated in their analysis of economic phenomena. Outlined by the Federal Reserve (2017) and the IMF (2024), macroeconomics examines the overall behavior and performance of the economy, focusing on broad factors such as business cycles, economic growth, inflation, and the effects of government policies. This field extends its analysis to the international level, reflecting global linkages through trade and investment flows.

Microeconomics, on the other hand, focuses on individual economic agents such as firms, households and consumers, and the decision-making processes that affect their production, pricing, and consumption. It examines how these decisions affect market conditions and resource allocation, and addresses issues like consumer behavior, firm production, market competition and economic welfare. Microeconomics is well known for its practical applications in areas such as business strategy and public policy, providing understanding of how particular economic choices affect broader market dynamics. (OpenStax, 2024)

The relationship between macro- and microeconomics is essential for understanding the economy. Macroeconomic conditions influence microeconomic decisions, and the condition of national economies, for example, affects business strategies and consumer spending. Similarly, the combination of microeconomic activities can affect macroeconomic outcomes such as national productivity and economic growth. Together, these areas form a complete picture of economic activity and highlight the importance of both broad economic policies and individual market behaviors in shaping economic stability and growth. Building on the understanding of macro- and microeconomics, the focus shifts to how the state of the broader economy is measured by individual macroeconomic indicators. (OpenStax, 2024)

5.1.1 Macroeconomic indicators

In economic analysis, economic indicators at the macro level aim to give clear information about the key areas where policy measures can be most effective. Zaman (2007) provides an overview on the macroeconomic indicators and divides the indicators into seven sub-categories, which are output-, social field-, fiscal sector-, monetary-, price level-, external sector-, and transitional country specific indicators.

The macroeconomic environment is primarily understood through output indicators such as gross domestic product (GDP), gross national product (GNP), and industrial-, agricultural-, and services output indicators. In the social field, specific macroeconomic variables are relevant, covering areas such as education, health, and the labor market. These reflect the social dimension of economic activity and are related to the well-being and stability of society. The health of public finances is examined through fiscal indicators, mainly taxation and public expenditure. Monetary policy indicators provide information on the state of monetary conditions, including interest rates, money supply and demand, exchange rates, inflation, and access to financing. (Zaman, 2007)

Zaman (2007) continues that examining the price level provides information on the cost of commodities, labor (wages) and capital (interest rates). In addition, the price of foreign currency (exchange rate) and trade prices are examined to see how they affect international competitiveness. The external sector is assessed through the balance of payments, in particular the current and capital account, and the foreign trade balance. In transition economies, indicators may also cover privatization, economic restructuring and adjustments resulting from the transition to a market economy. Together, these macroeconomic indicators provide a comprehensive view of economic performance and guide policy formulation, as well as lay the groundwork for the analysis in this paper to examine their linkage to M&A activity.

5.2 The data collection and transformation

The M&A data has been obtained from Bloomberg Terminal as it provides detailed and reliable financial information and has been widely used to conduct many M&A-related studies. Bloomberg also has a very comprehensive database of unlisted company transactions, which is important for this study because the overall M&A market in Europe is studied instead of just the listed market. Furthermore, the database provides accurate information on the details of each transaction, allowing for easier data processing and cleaning. For the macroeconomic indicators, the Statistical Office of the European Commission (Eurostat) was chosen because it is a highly reliable source of information, maintained by the central authority of the

European Commission. This ensures the data's accuracy, consistency, and official endorsement across the European Union, thus also enabling easier comparability. Additionally, after comparing different databases, Eurostat appeared to provide comprehensive datasets best suited to the specific needs of this research. These two sources will be used to provide a basis for analyzing how macroeconomic indicators influence or reflect M&A activity. All values are EUR denominated whether retrieved from Bloomberg or Eurostat. The following section details the process of data collection from both sources and presents the final datasets.

5.2.1 M&A data collection

As stated earlier, the M&A data was collected from the Bloomberg terminal. The initial selection criteria for the data included all completed M&A deals where the acquirer is European and a majority stake in the target company is acquired. The acquirer's geographical positioning is based on the Bloomberg terminal Country/Region selection criterion, which was set to cover "Eastern Europe" and "Western Europe". The acquiring countries are: Albania, Austria, Azerbaijan, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Georgia, Germany, Gibraltar, Greece, Guernsey, Hungary, Iceland, Ireland, Isle of Man, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Macedonia, Malta, Monaco, Montenegro, Netherlands, Norway, Poland, Portugal, Romania, Russia, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom. The target is not limited geographically and can be located anywhere in the world.

The time period for the collected transactions extended from January 1.1.2017 to 31.12.2022. It covers all completed M&A transactions announced during that time in Europe, totaling 39,371 transactions. This observation period was chosen to provide a view on the determinants of the M&A market before and during Covid-19 pandemic. Years 2017-2019 cover the time before Covid-19 and as the pandemic started in late 2019 in China and was declared as public health emergency of the whole world by WHO in January 2020, the time period representing Covid-19 is 2020-2022. This way the whole period covers six years, three of which are before introduction and three during the pandemic.

This is the raw data that was collected from Bloomberg, and the data is further processed in Microsoft Excel. All transactions whose value is not reported are removed, as well as transactions with a value of less than €1 million, in line with Moeller et al. (2005) and Choi and Jeon (2011), who used the \$1 million threshold. Transactions involving acquirer or target companies in the financial services or banking industries are also excluded as these companies

are subject to strict regulations and fundamentally differ from others. This brings the total to 9,131 transactions, which is the final transaction amount to be analyzed.

Table 2 M&A transaction data collection

M&A transaction data collection	
Deal type	M&A, where a majority stake is acquired
Deal status	Completed
Acquirer's geographical positioning	Europe: Albania, Austria, Azerbaijan, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Georgia, Germany, Gibraltar, Greece, Guernsey, Hungary, Iceland, Ireland, Isle of Man, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Macedonia, Malta, Monaco, Montenegro, Netherlands, Norway, Poland, Portugal, Romania, Russia, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom
Target's geographical positioning	Not limited geographically
Announcement period	1.1.2017 – 31.12.2022
Deal value	≥€1 million
Acquire and target industries	All excluding financial services and banking industries

After the data cleaning process, the priorities include analyzing the total number of M&A and their values, so that they can be compared with the macroeconomic indicators, which are reported on a monthly basis. To minimize the effect of outliers, winsorization is used with thresholds of 1% and 99%, which limits the range of the data to avoid bias in the analysis. This method adjusts the data set by cutting off the most extreme data points and setting all outliers to the closest value within the specified percentage limits. Once the daily data are winsorized, they are aggregated into monthly figures. At this stage, the total number of transactions for each month is calculated and the winsorized transaction values are summed to give a consolidated picture of the month's activity. The aggregation of monthly data allows for a clearer analysis of trends over time and balances out any variation in the daily data. The

process culminates in the creation of a comprehensive M&A dataset covering a total of six years or 72 months.

Figure 2 The final M&A data sample

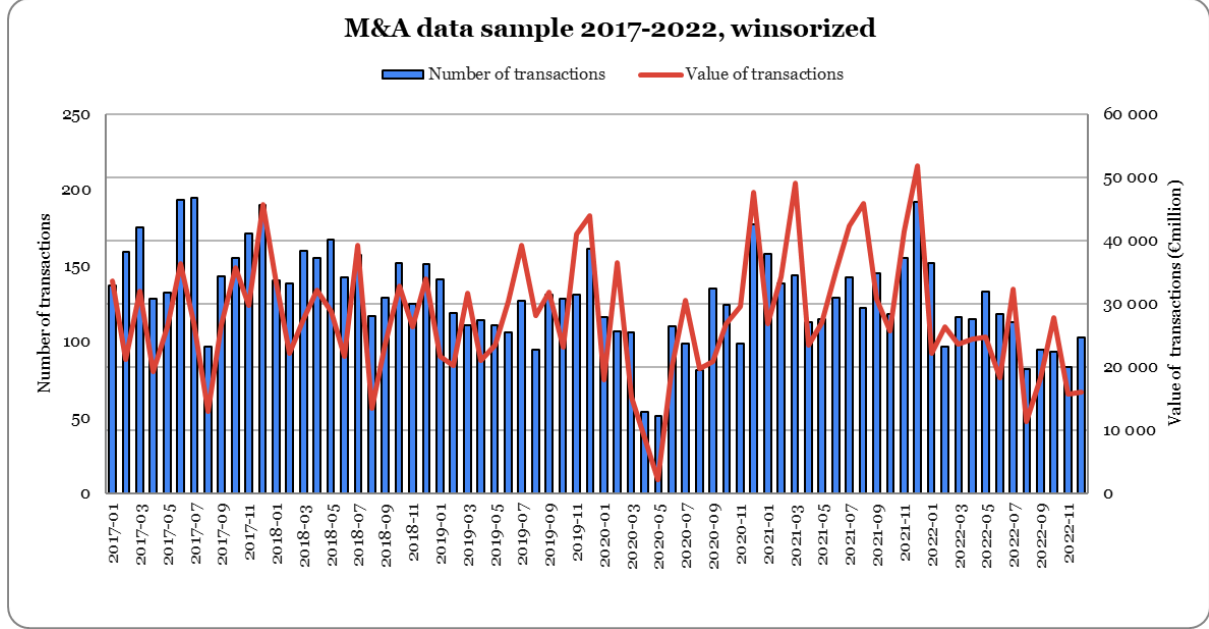


Figure 2 presents the final winsorized M&A data, from January 2017 until December 2022 and it shows that the number and value of transactions varies significantly from month to month. The highest number of transactions occurred in July 2017, with 195, and the lowest in May 2020, with 51. December 2021 marked the highest month in terms of transaction value, as it was around €51.83 billion, while the lowest monthly value was around €0.89 billion in April 2020.

The data show that the number of M&A transactions decreased significantly in early 2020, corresponding to the beginning of the Covid-19 pandemic. Over a six-year period, there is no clear trend in the number of M&A transactions, with activity fluctuating between high and low levels. The same is true for the value of transactions, with significant peaks and troughs but no consistent trend. Monthly changes in the number and value of transactions do not move consistently in the same direction, suggesting that transaction activity may be influenced by a number of different factors rather than by any single trend.

5.2.2 Macroeconomic data collection

In order to conduct a comprehensive analysis of the state of the economy, five key indicators were selected for the macroeconomic dataset, all of which were obtained from Eurostat, as mentioned above. The indicators were chosen because they are relevant and representative of

the economic climate in the Europe area, with a constant composition to ensure a consistent comparison. The indicators that were selected are: EU27 fixed composition 2020 Gross Domestic Product (GDP) at market prices, the 3-month Euribor interest rate, the EURO STOXX 600 equity index, the unemployment rate in Europe and the exchange rate between the US dollar and the euro.

As the GDP for the EU27 fixed composition in 2020 was used, it excludes two major countries from the M&A dataset, namely the U.K. and Russia. The U.K.'s GDP was retrieved from Eurostat, and Russia's GDP from the Federal Reserve Bank of St. Louis, with their respective average exchange rates for each quarter to enable conversion into euros. The GDPs of these two countries were then added to the EU27 fixed composition GDP at market prices to create a macroeconomic dataset comparable with the M&A data. GDP figures, which are normally reported quarterly, were converted to monthly figures to be consistent with other indicators. This was done by dividing the quarterly GDP by three to obtain an estimated monthly GDP value to be used for each month of each quarter. This approach allows for a more accurate monthly analysis and comparison with other monthly reported indicators.

The final dataset contains 72 monthly observations for each indicator for the period January 2017 to December 2022. The 3-month Euribor reflects the average interest rate at which euro area banks lend unsecured funds to other banks over a 3-month period, in real terms. The STOXX600 index, indexed to 100 in December 2016, serves as an indicator of the overall performance of European stock markets, tracking the stock performance of 600 large, mid, and small-cap companies across 17 European countries. The unemployment rate describes the average percentage of unemployed people in the labor force during a month, and the US dollar/euro exchange rate is calculated as the average of daily observations for the month, reflecting the strength, stability and attractiveness of the European markets as well as forecasting its economic performance.

Table 3 Macroeconomic indicator data collection

Macroeconomic indicator data collection	
Gross domestic product	Gross domestic product at market prices, quarterly (summed through period), EU27 fixed composition. The U.K. and Russia's figures at market prices added respectively
Interest rate	Real 3-month Euribor (Euro Interbank Offered Rate), average of observations through month

Stock market performance	EURO STOXX Equity Index (indexed 100=12/2016), average of observations through month
Unemployment rate	Unemployment rate, EU27 fixed composition, average of observations through month
Exchange rate	US dollar/euro (USD/EUR), average of observations through month

5.3 Variable description

After a comprehensive presentation of the data collection process and the data itself, the focus shifts to detailing and explaining the motives for the different variables chosen, which will be used in the empirical part of the study.

5.3.1 Dependent variables

The number and value of transactions are often considered the clearest measures of M&A activity. Choi & Jeon (2011) underscore the importance of these metrics in their study of U.S. M&A activity and its relationship with macroeconomic indicators and adopts those accordingly as its dependent variables. In this study, *Number of transactions* counts the announced deals that are successfully completed, and *Value of transactions* refers to the total value of these completed deals.

The third dependent variable to be used is *European target*, which measures the share of the number of transactions originating from Europe in the global context, as the target company is not limited geographically. It is calculated by dividing the number of European transactions (where the target company is in Europe) in each month by the total number of transactions in that month. This gives an indication of whether European companies prefer to acquire domestic over foreign companies. In the later sections, this is also denoted as intra-European M&A activity.

5.3.2 Independent variables

Based on the dependent variables, this study uses several independent variables to examine their predictive power for M&A transactions. These variables are selected based on their potential impact on the market dynamics of mergers and acquisitions and on previous studies.

GDP reflects overall economic health and market size and serves as an indicator of the economic environment in which M&A decisions are made. *GDP* has been used as an independent variable in several previous studies (Resende, 2008; Choi & Jeon, 2011; Uddin & Boateng, 2011). The 3-month *Euribor*, which represents the average interest rate at which banks in the Euro area lend to each other, reflects the cost of capital that can affect the financing of M&A transactions. Interest rates overall are also used widely in previous studies (Beckett, 1986; Choi & Jeon, 2011; Uddin & Boateng, 2011). The STOXX Europe 600 equity index, commonly referred to as *STOXX*, is an equity index that tracks the equity performance of 600 large, medium, and small cap companies in 17 European countries and serves as a measure of overall market sentiment. All of the earlier presented previous studies have included some stock market index in their analysis.

Unemployment rates are included to reflect economic sentiment from a labor market perspective, which can indirectly affect M&A. Interestingly, to my knowledge, the previous literature does not provide evidence of an analysis on the explanatory value of unemployment rates in M&A. The USD/EUR exchange rate provides information on the relative strength of the euro and can affect M&A activity through changes in purchasing power and investment attractiveness. Exchange rates were analyzed as an explanatory variable by Uddin and Boateng (2011), and Ibrahim and Raji (2017) used the USD exchange rate to define the relative price of domestic currency to USD, which motivates the use of the USD/EUR exchange rate in this study.

To account for the time-lag impact of these economic indicators, new variables are created by adding a one-month lag to each of the original variables, denoted by "*_lag*", reflecting the possible delayed impact of the economic situation on M&A activity. The variables are *GDP_lag*, *Euribor_lag*, *STOXX_lag*, *Unemployment_lag*, and *Exchange rate_lag*. This allows an analysis of whether recent past economic indicators affect M&A activity. Finally, the variable *Covid* is used as a dummy variable to examine the impact of the Covid-19 pandemic on M&A activity, differentiating between the period before and during the pandemic.

5.3.3 Natural logarithm of the variables

In regression analysis, it is common to take the natural logarithm (Ln) of the variables. Several previous studies in the field of M&A have used log-transformed variables in regression analyses. Ibrahim and Raji (2017) employed the natural logarithm for all variables in their study on cross-border M&A and the role of macroeconomic indicators. Similarly, Uddin and Boateng (2011) applied natural logarithm transformations in their study, which focused on the UK. They state that transforming the variables helps in examining the response of dependent

variable for a given change in one independent variable holding other independent variables constant.

This approach has several advantages. It helps address the skewness that is often present in economic data, with a few large values distorting the overall picture. By using logarithms, data distributions become more symmetrical, contributing to the statistical models' requirement for normality. Moreover, when analyzing changes, the coefficients of logarithmically transformed variables show the percentage change in the outcome (dependent variable) for a one percent change in the predictor (independent variable). This can be useful because ratios and relative changes are often more significant than absolute numbers.

Also, economic variables frequently interact in a way that is multiplicative rather than linear. Transforming these variables to logged variables turns these multiplicative relationships into additive ones, which makes the analysis and interpretation of their interactions simpler. Logarithmic transformation is also valuable for comparing variables that vary widely in magnitude, such as GDP, which can be much larger than other variables. This transformation ensures that larger values do not influence the model in an unwanted way.

Finally, using the natural logarithm can lead to better model accuracy. It tends to stabilize the variance of residuals, which is the variability of the observed dependent variable values that cannot be explained by the model, referred to as homoscedasticity. Thus, in this study, logarithmic transformations of all the variables are used to enhance the analysis, providing clear insights into how changes in macroeconomic indicators relate to changes in M&A activity.

5.4 Descriptive statistics

The table below presents the descriptive statistics for the three dependent and six independent variables including the Covid-19 dummy variable.

Table 4 Descriptive statistics

Variable	Observations	Average	Median	Std. Dev	Min	Max	Skewness	Kurtosis
Number	72	129,35	128,50	30,14	51,00	195,00	-0,006	0,034
Value	72	27,89	26,94	9,78	2,14	51,83	0,193	-0,005
European_target	72	0,72	0,72	0,05	0,61	0,84	0,033	-0,332
GDP	72	1431,42	1414,70	83,39	1222,23	1608,23	0,163	0,466
Euribor	72	-0,03	-0,02	0,03	-0,09	0,00	-1,371	0,524
STOXX	72	114,26	111,91	11,07	90,06	139,76	0,438	-0,056
Unemployment	72	0,07	0,07	0,01	0,06	0,09	0,286	-0,764
Exchange_rate	72	1,13	1,14	0,06	0,98	1,23	-0,494	-0,227
Covid	72	0,50	0,50	0,50	0,00	1,00	0,000	-2,028

Note: Value and GDP in €billion

The data consists of 72 observations for each variable covering a six-year period. The variable *Number*, describing the number of M&A transactions, has an average of 129.35 and a median of 128.50, indicating that the distribution is symmetric around the central trend, confirmed by a skewness value close to zero (-0.006). The *Value* variable describing the total value of transactions in billions of euros has an average of 27.89 billion and a standard deviation of 9.78 billion, suggesting a moderate variation from the mean. The average and median of the *European target* variable, which describes the share of intra-European transactions, are consistently 0.72, with little variation, with a standard deviation of 0.05. As the variable describes European M&A transactions, it means that the share of European transactions is relatively stable over the observed months during 2017-2022.

For the macroeconomic indicators, the average value of *GDP* expressed in billions of euros is 1 431.42, with a variation between the minimum and maximum values, reflected in a standard deviation of 83.39 billion. The skewness is near-zero (0.163), which suggests that the GDP distribution is symmetrical, and that there are as many periods of above-average growth as there are below-average. The kurtosis (0.466), however, indicates a relatively light-tailed distribution compared to a normal distribution, suggesting fewer outliers or extreme values in GDP data than would be expected in a normal distribution. The *Euribor*, with an average value of -0.03, highlights the negative interest rate environment during the observed period. Its skewness of -1.371 further suggests that the distribution is clustered closer to zero with a tail pointing to occasional periods of significantly lower rates. The *STOXX* index, with an average of 114.26, reflects the overall performance of the stock market, with a distribution skewed slightly to the right (0.438). In contrast, the unemployment rate, with a narrow range of 0.06-0.09, is positively skewed (0.286), which may indicate a clustering of lower unemployment rates over most months. The average value of the USD/EUR exchange rate is 1.13, close to its median, suggesting a relatively balanced distribution of exchange rates over the period. However, the negative skewness (-0.494) indicates that the exchange rate tends to lean towards higher values, reflecting a stronger euro against the US dollar.

5.5 Correlation matrix

The table below presents the correlation matrix for the three dependent and six independent variables including the Covid-19 dummy variable.

Table 5 Correlation matrix

	Number	Value	European_target	GDP	Euribor	STOXX	Unemployment	Exchange rate	Covid
Number	1								
Value	0,629	1							
European_target	-0,045	-0,309	1						
GDP	-0,046	0,116	-0,115	1					
Euribor	0,185	0,131	-0,052	-0,849	1				
STOXX	0,210	0,398	-0,254	0,563	-0,459	1			
Unemployment	0,428	0,106	0,104	-0,700	0,659	-0,339	1		
Exchange rate	0,392	0,322	-0,150	-0,436	0,608	0,160	0,449	1	
Covid	-0,396	-0,094	-0,150	0,268	-0,418	0,428	-0,487	-0,148	1

The correlation matrix of the data shows that there are varying degrees of linear relationships between the variables. The positive correlation (0.629) between the *Number* and *Value* of transactions suggests that as one variable increases, so does the other. In contrast, the *European target* variable has a small negative correlation (-0.045) with the *Number* of transactions and a more significant negative correlation (-0.309) with the *Value* of transactions. This suggests that a higher proportion of European targets is associated with a slight decrease in the number and a more noticeable decrease in the value of transactions.

GDP has a weak correlation with the *Number* of transactions (-0.046), while it is slightly stronger and in the opposite direction with the *Value* of transactions (0.116). *Euribor* is slightly positively correlated with the *Number* of transactions (0.185) and with the *Value* (0.131). The *STOXX* index has a moderate positive correlation with the *Value* of transactions (0.398) and a weaker but still positive correlation with the *Number* of transactions (0.210). The *Unemployment* rate is more strongly correlated with the *Number* of transactions (0.428) than with the *Value* (0.106).

The *USD/EUR* exchange rate is positively correlated with the *Number* and *Value* of transactions (0.392 and 0.322). Interestingly, there is a negative correlation of -0.150 of *European target* and the *USD/EUR*, which implies that when the euro strengthens against the dollar (which would cause the *USD/EUR* rate to decrease), there is a slight tendency for the proportion of M&A transactions involving European targets to increase. Conversely, when the euro weakens (and the *USD/EUR* rate increases), the proportion involving European targets may see a slight decrease.

The Covid-19 variable has a moderate negative correlation (-0.396) with the *Number* of transactions, suggesting a possible decrease in the number of transactions during the pandemic. However, its correlation with the *Value* of transactions is less significant, -0.094. These correlations provide a preliminary idea of how different macroeconomic factors, and the Covid-19 may be related to M&A activity.

6 METHODOLOGY

This section outlines the methodology employed in this study. First, the research hypothesis will be formulated based on theory and existing literature, followed by the corresponding statistical hypothesis. Regression analysis will then be conducted to test the explanatory power of the chosen independent variables on the dependent variable. Finally, the model diagnostics will be presented and analyzed to ensure the validity of the results.

6.1 Hypotheses formulation

This study uses a multi-stage approach to examine the impact of macroeconomic indicators and the Covid-19 pandemic on M&A activity in Europe, by testing the chosen macroeconomic variables separately. Building on the established literature, the theoretical framework presented and the purpose of this study, nine research hypotheses are formulated. These hypotheses primarily assess whether different macroeconomic indicators can explain the observed M&A activity, where the acquiring company is European. To test these hypotheses statistically, four corresponding statistical hypotheses will be formulated.

Higher GDP growth is typically a sign of a strong economic environment, which boosts business confidence and investment, which in turn drives M&A activity. This positive relationship between GDP growth and M&A activity is supported by economies of scale and increased market demand that arise as the economy grows (Stoian and Filippaios, 2008). As GDP grows, firms are better able to finance M&A due to improved financial performance, but they are also motivated to diversify and manage risk by acquiring other firms both domestically and internationally (Globerman and Shapiro, 1999; Uddin and Boateng, 2011). While Billington (1999) and Caves (1989) argue that GDP growth can also lead to an increase in local acquisitions, which discourages international investment, the general trend suggests that with economic growth, firms have both the means and the confidence to actively pursue mergers and acquisitions and seek to exploit new opportunities offered by thriving markets.

H1: The relationship between GDP growth and M&A activity is positive.

Interest rates have a significant impact on cross-border investments, especially M&A. As Moeller and Schlingemann (2005) point out, financing costs play a crucial role in M&A decisions, and higher interest rates may discourage such transactions. This view is supported by Tolentino (2010), who links lower domestic interest rates to capital availability, which facilitates foreign investment to improve profitability. Empirical research further highlights this relationship. For example, Aliber (1970) (as cited in Boateng et al., 2014) and Barrell and

Pain (1996) find that lower financing costs encourage firms to undertake foreign direct investment, including M&A. A similar argument is made by Pablo (2009) and Forssbaeck and Oxelheim (2008), who argue that cost and access to capital is a major factor for the ability to undertake cross-border M&A, which is in line with Uddin and Boateng (2011), who suggest lower cost of capital increase the likelihood that a firm will make outward M&A due to increased financing synergies. However, Forssbaeck and Oxelheim (2008) also point out that in efficient markets all firms have equal access to capital at similar costs, which may level conditions regardless of domestic interest rates. Considering the previous literature, the following hypothesis is formulated:

H2: The relationship between higher 3-month Euribor rates, indicating higher short-term interest rates, and M&A activity is negative.

Early studies by Weston (1953) and Nelson (1959) show that stock prices significantly affect M&A activities. Specifically, Nelson observed a positive relationship between stock price fluctuations and merger activities in the U.S. during 1895-1920. Researchers like Shleifer and Vishny (2003) and Baker et al. (2009) supported the idea that stock prices impact M&A dynamics using the overvaluation hypothesis. They noted that during stock market booms, some companies' shares may become overvalued. To protect shareholders' interests against a possible fall in share prices, managers could use overvalued shares to acquire real assets through acquisitions. Baker et al. (2009) also suggested that overvalued domestic shares could be leveraged to acquire foreign firms, which they referred to as the "cheap financial capital hypothesis". In addition, Vasconcellos and Kish (1998) found that the weaker US stock market compared to foreign markets may encourage foreign acquisitions of U.S. firms. Based on the previous studies, the following hypothesis is formulated:

H3: The relationship between increases in the STOXX600 index and M&A activity is positive.

Given the limited direct empirical evidence on the link between unemployment rates and firm-level trading activity, the following hypothesis aims to explore an area that has been hardly studied. The hypothesis is based on the theoretical argument that high unemployment rates may be an indicator of broader economic distress, which may affect firms' strategies, including M&A activity. Higher unemployment tends to indicate a reduction in consumer purchasing power, which can lead to lower revenues for firms and potentially lower profits across sectors. Economic contraction may discourage firms from expanding through acquisitions due to increased financial risk and uncertainty about future returns. Moreover, in the context of high

unemployment, firms may prioritize cost-cutting and efficiency over growth strategies involving M&A. By using these arguments, the following hypothesis is formulated:

H4: The relationship between higher unemployment rates and M&A activity is negative.

According to the overvaluation hypothesis, mergers and acquisitions are often the result of valuation differences between the acquirer and the target companies, which are significantly affected by exchange rates (Boateng et al., 2014). Erel et al. (2012) find that international valuation differences can be the cause of cross-border M&A, especially when the domestic currency of the firm strengthens, making foreign firms more financially attractive and more accessible. This view is supported by Vasconcellos and Kish (1996), who highlight the crucial importance of the relative strength of the currency in M&A decisions. Exchange rate fluctuations affect not only the initial acquisition costs but also the broader economic dynamics of the transaction, including financial and operational management costs post-acquisition (Weston and Jawien, 1999). For European firms, a stronger domestic currency facilitates acquisitions abroad by effectively lowering the price of foreign assets. Studies such as Froot and Stein (1991) and Goergen and Renneboog (2004) support this view and suggest that firms from countries with stronger currencies tend to be more active acquirers in international markets. Therefore, the following two hypotheses can be formulated:

H5: The relationship between a strengthening euro (USD/EUR exchange rate decreases) and the overall M&A activity is positive.

H6: The relationship between a strengthening euro (USD/EUR exchange rate decreases) and the intra-European M&A activity is negative.

The use of lag to examine the relationship between macroeconomic variables and M&A activity can effectively reflect the timing of firms' decision-making. M&A strategies are usually not based on immediate economic changes but go through a strategic assessment and financial planning process, which naturally introduces a lag. Economic changes, such as shifts in Euribor interest rates, affect the cost of financing acquisitions, but it takes time to implement these changes in companies' financing strategies. Empirical evidence supports this approach (Boateng et al., 2014; Uddin & Boateng, 2011; Bhagwat et al., 2016), showing that firms' reactions to financial changes are generally observed with a short lag, corresponding to the time required by an organization's decision-making processes. Economic information is often

available with a lag, and firms act based on the latest available information. Therefore, the following hypothesis is formulated:

H7: Lagged economic indicators are more significant predictors of M&A activity than original indicators.

It can be argued that the health of the macroeconomic environment plays a crucial role in shaping investors' behavior and decisions. When European economies show positive developments in macroeconomic indicators, such as strong GDP growth and low unemployment rates, the financial performance and prospects of companies tend to improve in these economies. These favorable conditions not only improve corporate earnings but also increase growth potential, making companies more attractive targets for acquisitions. In addition, favorable economic conditions can significantly reduce the risks associated with investment. Investors and acquiring companies are more willing to undertake M&A when they are confident about the financial stability of the target market, which promises more predictable returns on investment. This is particularly important in Europe, where interconnected economies mean that positive changes in one country can boost investor confidence across the region. In addition, companies operating in thriving economies often have access to better financing options at lower interest rates, making them even more attractive as potential acquisition targets. Given these dynamics, the following hypothesis is formulated:

H8: The relationship between intra-European M&A transactions and positive shifts in European macroeconomic indicators is positive.

The Covid-19 pandemic has undoubtedly affected global economic conditions, which may have affected mergers and acquisitions due to changes in market conditions, the regulatory framework and increased economic uncertainty. External shocks may lead firms to adjust their strategies, which may affect their M&A activity. For example, a pandemic could lead firms to re-evaluate their business models in response to changes in consumer behavior or supply chain disruptions. Andrade et al. (2001) and Mulherin and Boone (2000) have identified that merger activities tend to cluster, and this clustering is influenced by shocks in technology, economy, and regulation. The new regulatory measures and general economic uncertainty may change the strategic and financial calculations underlying acquisition decisions. Choi and Jeon (2011) find that economic growth cycles increase M&A activity and Cerrato et al. (2016) found that firms focus more on core business activities and domestic markets during

downturns and reduce diversification and international acquisitions, which could lead to lower M&A activity. Therefore, the following hypothesis is formulated:

H9: The Covid-19 pandemic affects the M&A activity in all regressions.

Table 6 Summary of research hypotheses

Hypothesis 1	The relationship between GDP growth and M&A activity is positive.
Hypothesis 2	The relationship between higher 3-month Euribor rates, indicating higher short-term interest rates, and M&A activity is negative.
Hypothesis 3	The relationship between increases in the STOXX600 index and M&A activity is positive.
Hypothesis 4	The relationship between higher unemployment rates and M&A activity is negative.
Hypothesis 5	The relationship between a strengthening euro (USD/EUR exchange rate decreases) and the overall M&A activity is positive.
Hypothesis 6	The relationship between a strengthening euro (USD/EUR exchange rate decreases) and the intra-European M&A activity is negative.
Hypothesis 7	Lagged economic indicators are more significant predictors of M&A activity than original indicators.
Hypothesis 8	The relationship between intra-European M&A transactions and positive shifts in European macroeconomic indicators is positive.
Hypothesis 9	The Covid-19 pandemic affects the M&A activity in all regressions.

6.1.1 Statistical hypotheses

Statistical tests enable the results to be examined to determine their significance. At predefined levels of statistical significance, results can be considered either significant or not significant. In line with the purpose of the study, four statistical hypotheses have been formulated to provide statistical validation of the interpretation of the results. Although there is a broader set of research hypotheses presented earlier, each hypothesis has not been defined separately as a statistical hypothesis in this section. Instead, the statistical results for each variable are presented in detail in the Results section, which provides a deeper analysis of the implications and nuances of the explanatory power of the variables. The primary four statistical hypotheses considered here are as follows:

Statistical hypothesis 1:

H₁₀: European macroeconomic indicators have no explanatory power over the variation in M&A activity.

H₁₁: European macroeconomic indicators have explanatory power over the variation in M&A activity.

Statistical hypothesis 2:

H₂₀: European macroeconomic indicators do not explain the tendency to acquire companies from Europe.

H₂₁: European macroeconomic indicators explain the tendency to acquire companies from Europe.

Statistical hypothesis 3:

H₃₀: The Covid-19 pandemic has no impact on M&A activity.

H₃₁: The Covid-19 pandemic has an impact on M&A activity.

Statistical hypothesis 4:

H₄₀: The Covid-19 pandemic has no impact on the intra-European M&A activity.

H₄₁: The Covid-19 pandemic has a significant impact on the intra-European M&A activity.

6.2 Multiple regression analysis

In examining the influence of macroeconomic variables on M&A activity, this study utilizes multiple linear regression (MLR), a widely used statistical method in finance for estimating the relationship between one dependent variable and multiple independent variables. The underlying econometrical model for the MLR is:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_n x_{ni} + \varepsilon_i$$

, where y_i is the dependent variable, β_0 is the intercept term, $\beta_1, \beta_2, \dots, \beta_n$ the coefficients of the independent variables, $x_1, x_2, \dots, x_n$ are the independent variables and ε_i is the error term.

MLR employs the Ordinary Least Squares (OLS) method to find the line of best fit that minimizes the sum of the squared differences between the observed values and the values

predicted by the model (Chatterjee & Simonoff, 2013). The objective function that the OLS regression minimizes is:

$$\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_n x_{ni}))^2$$

The regression will use the natural logarithm of all variables to capture elasticity when examining the results (Uddin & Boateng, 2011). The number and value of M&A deals serve as the first two dependent variables, whereas the third represents the ratio of deals executed in Europe. The main independent variables will be the natural logarithms of macroeconomic indicators, including GDP figures, Euribor rates, STOXX index values, unemployment rates, and USD/EUR exchange rates. Additionally, the model will include a dummy variable for the Covid-19 period to capture any effects related to the pandemic. The decision to incorporate these specific variables draws from the empirical framework used by Chiriac (2021) and is further supported by related studies including Ibrahim and Raji (2017), Choi and Jeon (2011), and Uddin and Boateng (2011). The regression models presented in the equations provide a detailed understanding of the dynamics between economic indicators and M&A activity. Regressions (1-4) aim to investigate the following research questions:

How do European macroeconomic indicators explain M&A activity?

Does the Covid-19 pandemic have an impact on the M&A activity?

Regression 1:

Ln(Number) =

$$\beta_0 + \beta_1 \text{Ln(GDP)}_i + \beta_2 \text{Ln(Euribor)}_i + \beta_3 \text{Ln(STOXX)}_i \\ + \beta_4 \text{Ln(Unemployment)}_i + \beta_5 \text{Ln(Exchange_rate)}_i + \beta_6 \text{Covid}_i + \varepsilon_i$$

Regression 2 (lag):

Ln(Number) =

$$\beta_0 + \beta_1 \text{Ln(GDP_lag)}_i + \beta_2 \text{Ln(Euribor_lag)}_i + \beta_3 \text{Ln(STOXX_lag)}_i \\ + \beta_4 \text{Ln(Unemployment_lag)}_i + \beta_5 \text{Ln(Exchange_rate_lag)}_i + \beta_6 \text{Covid}_i + \varepsilon_i$$

Regression 3:

Ln(Value) =

$$\beta_0 + \beta_1 \text{Ln(GDP)}_i + \beta_2 \text{Ln(Euribor)}_i + \beta_3 \text{Ln(STOXX)}_i \\ + \beta_4 \text{Ln(Unemployment)}_i + \beta_5 \text{Ln(Exchange_rate)}_i + \beta_6 \text{Covid}_i + \varepsilon_i$$

Regression 4 (lag):

$\text{Ln}(\text{Value}) =$

$$\beta_0 + \beta_1 \text{Ln}(\text{GDP_lag})_i + \beta_2 \text{Ln}(\text{Euribor_lag})_i + \beta_3 \text{Ln}(\text{STOXX_lag})_i \\ + \beta_4 \text{Ln}(\text{Unemployment_lag})_i + \beta_5 \text{Ln}(\text{Exchange_rate_lag})_i + \beta_6 \text{Covid}_i + \varepsilon_i$$

Regressions (5-6) aim to investigate the following two research questions:

How do the European macroeconomic indicators explain the tendency to acquire companies from Europe?

Does the Covid-19 pandemic have an impact on the intra-European M&A activity?

Regression 5:

$\text{Ln}(\text{European_target}) =$

$$\beta_0 + \beta_1 \text{Ln}(\text{GDP})_i + \beta_2 \text{Ln}(\text{Euribor})_i + \beta_3 \text{Ln}(\text{STOXX})_i \\ + \beta_4 \text{Ln}(\text{Unemployment})_i + \beta_5 \text{Ln}(\text{Exchange_rate})_i + \beta_6 \text{Covid}_i + \varepsilon_i$$

Regression 6 (lag):

$\text{Ln}(\text{European_target}) =$

$$\beta_0 + \beta_1 \text{Ln}(\text{GDP_lag})_i + \beta_2 \text{Ln}(\text{Euribor_lag})_i + \beta_3 \text{Ln}(\text{STOXX_lag})_i \\ + \beta_4 \text{Ln}(\text{Unemployment_lag})_i + \beta_5 \text{Ln}(\text{Exchange_rate_lag})_i + \beta_6 \text{Covid}_i + \varepsilon_i$$

6.3 Model diagnostics

In statistical models, particularly when using the OLS estimation method, several assumptions are necessary to validate the accuracy and reliability of the model's results. Firstly, it is assumed that the error terms in the model follow a normal distribution. In addition, homoscedasticity, where the variation between error terms remains constant across levels of independent variables, is crucial to ensure that the predictive power remains consistent. Multicollinearity among independent variables should be avoided to prevent unstable parameter estimates and interpretational difficulties. The absence of autocorrelation (uncorrelated error terms) is also essential to maintain the independence assumption that is necessary for unbiased parameter estimates. Furthermore, error terms should not be correlated with independent variables, which ensures that errors are not systematically influenced by those values. In financial research, these assumptions are important because the noise and complexity of the financial data present challenges, such as outliers and abnormal distributions, which need to be considered to maintain integrity in the models. This chapter

will systematically review the diagnostic tests conducted on our models. These tests are important to confirm the validity of the model results and their interpretations. (Brooks, 2019)

6.3.1 Normal distribution in the error terms

The normality of the error terms is a fundamental assumption of regression analysis, as stated earlier. To verify this assumption, the Jarque-Bera (JB) test is commonly used, where a high p-value usually indicates that the distribution of error terms does not deviate significantly from the normal distribution.

In this study, the JB test was used in all the regressions to test the normality of the residuals. For models 1, 2, 5 and 6, the p-values of the JB test were 0.766, 0.805, 0.771 and 0.937, respectively, these values are well above common thresholds of statistical significance, suggesting that the null hypothesis of normally distributed errors cannot be rejected for these models. Thus, these models satisfy the assumption of normality, which confirms the reliability of statistical tests.

In contrast, the low p-values in models 3 and 4 initially suggest a deviation from the normal distribution of error terms. However, given the large sample size of well above 30 observations, the Central Limit Theorem mitigates this issue by ensuring that the OLS estimators are normally distributed. Therefore, the normality assumption's violation does not significantly affect our regression coefficients' estimates, allowing for reliable statistical analysis in these models.

6.3.2 Multicollinearity

According to Wooldridge (2013), high correlation between two or more independent variables is called multicollinearity. Multicollinearity is tested in regression models to ensure that independent variables are not strongly linearly dependent, which can compromise the interpretation of the regression coefficients. A correlation matrix showing the correlations between the variables is presented in Table 5, and it reveals no concerning issues. Another way of examining multicollinearity is the Variance Inflation Factor (VIF) test. The values from the test provide a concrete measure of multicollinearity, with values exceeding 5-10 typically indicating a cause for concern. (Kim, 2019)

In the provided VIF results for models in Appendix 1 without lagged variables and with lagged variables, all VIF values fall below 6, which is generally considered acceptable and indicative of minimal multicollinearity. This suggests that the independent variables in both models are

sufficiently independent of each other. Therefore, the regression results for these models are likely to be robust revealing no multicollinearity issues.

6.3.3 Heteroscedasticity

Heteroscedasticity refers to a situation where the variance of the error terms varies across observations, which can lead to inefficient estimates. The Breusch-Pagan (BP) test is a popular method for detecting heteroscedasticity, with a high p-value suggesting homoscedasticity (constant variance). (Wooldridge, 2013)

The results for the BP tests can be found in the results summary table. Models 1, 2, 5, and 6 show p-values of 0.352, 0.240, 0.369, and 0.262, respectively. These values are above thresholds that are considered significant, indicating that there is no evidence of heteroscedasticity within these models. This suggests that the assumption of constant variance in the error terms is valid in these regressions, which confirms the standard errors and test statistics obtained from them. However, models 3 and 4 have p-values of 0.003 and 0.000, respectively, indicating the presence of heteroscedasticity. To address the heteroscedasticity, robust standard errors have been used in those two models. This approach corrects for the varying variance and ensures that the statistical inference remains valid.

6.3.4 Autocorrelation

Autocorrelation, or serial correlation, occurs when the error terms of a regression model are not independent of each other between different time points (Wooldridge, 2013). It can be commonly found in time series data, where observations are collected over time at regular intervals, which is the case in this study. The Durbin-Watson (DW) test is used to detect autocorrelation, with a value close to 2 suggesting no autocorrelation, values approaching 0 indicating positive autocorrelation, and values toward 4 indicating negative autocorrelation.

The DW test results for the regressions can be found in the results summary table. Despite slightly lower p-values in models 1 and 2 compared to the others, none of them showed statistically significant evidence of autocorrelation. Therefore, it can be concluded that no true autocorrelation was detected for any of the models based on the DW tests.

7 EMPIRICAL RESULTS

In this section, the empirical results of the study are presented, organized into four main parts. Firstly, the impact of macroeconomic indicators on the number of transactions is examined. Next, the influence of these indicators on the value of transactions is analyzed. Subsequently, the effects of macroeconomic indicators specifically on intra-European transactions are investigated. Finally, the results regarding the impact of the Covid-19 pandemic on M&A activity are presented.

Chapter 6 presented in detail the models that were used to run the six regression models with three different dependent variables: number of transactions, value of transactions and European target. In the tables presenting the results, the coefficients are listed on the same row as the variables, with the t-statistics shown below them in parenthesis. The significance levels (p-values) are indicated by asterisks: *** indicates 1% significance level, ** 5% significance level and * 10% significance level. As previously discussed in Chapter 6, models 3 and 4 exhibit heteroscedasticity. To address this issue and ensure the statistical results are reliable, robust standard errors have been used in these specific models. The remaining models do not present any concerns, and their results can be accepted as presented originally.

7.1 Macroeconomic indicators' effect on the number of transactions

7.1.1 *Normal independent variables*

The regression results for the number of M&A transactions can be seen in Table 7, revealing several statistically significant relationships. The model intercept is significantly negative at the 5% level, with a value of -29.857, suggesting a negative effect on the number of transactions when macroeconomic indicators are not taken into account. However, this scenario is theoretical rather than realistic, as it would suggest a negative expected number of M&A transactions. Moreover, it assumes that certain independent variables, such as GDP, could be zero, which is not feasible in reality. The GDP shows a positive relationship with the number of transactions. Specifically, the model estimates a coefficient of 1.990, which is significant at the 5% level. This implies that a 1% increase in GDP is expected to result in a 1.99% increase in the number of M&A.

Conversely, the Euribor rate, which represents the interest rate at which European banks lend to one another, does not show a significant effect on the number of transactions, with a coefficient of -0.560. The STOXX index, which is a stock index representing companies in Europe, has a positive coefficient of 0.774, significant at the 5% level, indicating that better stock market performance is associated with a rise in M&A transactions. Interestingly, the

coefficient for unemployment is positive at 19.194 and highly significant at the 1% level. This suggests that higher unemployment rates may be associated with an increased number of M&A transactions, which may reflect market consolidation during economic downturns or other indirect economic factors. The exchange rate, representing the USD/EUR, has a statistically significant coefficient of 2.111. The significant coefficient suggests a strong and positive elasticity between the logged USD/EUR exchange rate and the logged number of M&A transactions. Specifically, it implies that a 1% increase in the USD/EUR exchange rate results in an approximately 2.11% increase in the number of M&A transactions.

In terms of model diagnostics, the p-value is essentially zero, confirming the overall statistical significance of the model. The R-squared value of 0.538 indicates that the model explains over half of the variance in the number of M&A transactions. The adjusted R-squared, which adjusts for the number of independent variables in the model, is somewhat lower at 0.495, yet it still indicates that a considerable proportion of the variability in the dependent variable is explained for by the model.

7.1.2 Lagged independent variables

The second regression with lagged macroeconomic indicators as independent variables reveals an improved fit for explaining the number of M&A transactions, as R-squared value of 0.557 increases from the 0.538 of the initial model, and an adjusted R-squared of 0.515, up from 0.495. These differences in the degree of explanation between the models suggest that the model focusing on past economic conditions marginally outperforms the one with the latest indicators.

A significant finding in this regression is the coefficient of the lagged STOXX index, which is 1.220 and highly significant at the 1% level, indicating a strong statistical relationship. This reflects the impact of stock market performance from the previous period on current M&A activity, a relationship stronger than what was observed with the same period STOXX index in the first model.

Additionally, the lagged unemployment rate maintains its positive significance with a coefficient of 18.242, also at the 1% level. This strengthens the view that the M&A market may react to changes in the unemployment rate with a temporal delay, potentially due to the strategic considerations that companies undertake in response to the publishing of these indicators.

It is notable that both GDP and the exchange rate in their lagged forms do not show significant effects on the number of M&A transactions. The lagged GDP's coefficient is 0.834, lacking statistical significance, which contrasts with the positive and significant coefficient found in the non-lagged model. The exchange rate, with a coefficient of 1.401, aligns with its non-lagged analysis in showing a positive relationship, but it lacks statistical significance. The consistency in the non-significance of the Euribor rate, both in its original and lagged forms, indicates that it may not be a determinant factor for the number of M&A transactions during the study period.

Table 7 Results – Number of transactions

The results illustrate the relationship between the number of M&A transactions and economic indicators, including GDP, Euribor, STOXX, Unemployment, Exchange rate, and the impact of Covid. The analysis is conducted using two models: one considering current values of the indicators and the other using lagged values (1-month lag). Each model includes an intercept term. The results are presented with coefficients and their respective t-values in parentheses. Statistical significance is indicated by *, **, and *** representing p-values of 0.1, 0.05, and 0.01, respectively. The bottom panel of the table shows the model p-value, residual standard error, R-squared, and adjusted R-squared values.

	(1)		(2)
GDP	1,990** (2,248)	GDP_lag	0,834 (0,977)
Euribor	-0,560 (-0,293)	Euribor_lag	0,137 (0,075)
STOXX	0,774** (2,062)	STOXX_lag	1,220*** (3,263)
Unemployment	19,194*** (3,646)	Unemployment_lag	18,242*** (3,487)
Exchange rate	2,111* (1,671)	Exchange rate_lag	1,401 (1,123)
Covid	-0,164*** (-2,769)	Covid	-0,187*** (-3,234)
Intercept	-29,857** (-2,501)	Intercept	-14,994 (-1,301)
Model, p-value	0,000	Model, p-value	0,000
Residual Std. Error	0,179	Residual Std. Error	0,177
R-squared	0,538	R-squared	0,557
Adjusted R-squared	0,495	Adjusted R-squared	0,515

7.2 Macroeconomic indicators' effect on the value of transactions

7.2.1 Normal independent variables

In the third model analyzing the value of transactions, the regression shows significant relationships between the value of M&A transactions and several macroeconomic indicators. The negative intercept, at a value of -94.863 and significant at the 5% level, suggests that at

the average levels of the macroeconomic indicators, a lower value of M&A transactions is expected. However, as explained earlier, this interpretation should be taken with caution.

The first significant finding from this model is the positive relationship between GDP and the value of transactions. With a coefficient of 6.725, the implication is that a 1% increase in GDP is associated with a 6.73% increase in the value of transactions. This reflects the conventional expectation that stronger economic output is conducive to higher transaction values. The Euribor rate is another indicator with a strong influence, showing a coefficient of 11.309, significant at the 1% level. This high coefficient indicates that an increase in the Euribor rate is linked to a considerable rise in the value of transactions, suggesting that higher interest rates correlate with increased transaction values. The stock market performance, as measured by the STOXX index, also shows a positive correlation with the value of M&A transactions, with a coefficient of 1.669 and 1% significance level. This relationship suggests the potential impact of stock market health on M&A activity and strong valuation levels, perhaps reflecting investor confidence or availability of capital for transactions.

Interestingly, this model also reveals a positive relationship between the unemployment rate and the M&A activity in terms of value of transactions, with a coefficient of 23.092, significant at the 10% level. This could further strengthen the idea that increased unemployment correlates with higher M&A transaction values, possibly due to industry consolidation in times of labor market stress as noted previously. In contrast, the exchange rate does not show a statistically significant effect on the value of transactions in this model. This could be a result of a more complex nature of exchange rate fluctuations' impact M&A activity.

The R-squared of the model is 0.462, suggesting that close to half of the variability in the value of transactions is accounted for by the included macroeconomic indicators. The adjusted R-squared, at 0.412, provides a slightly lower value. This reveals that the model better explains the variation in the number of transactions compared to the value variable when considering the M&A activity with the indicators. The overall significance of the model is confirmed by a p-value of 0.000, which indicates that the macroeconomic factors included in the model together have a statistically significant effect on the value of M&A transactions.

7.2.2 Lagged independent variables

The fourth model, using lagged macroeconomic indicators to explain the value of M&A transactions, suggests different dynamics compared to the immediate effects captured in the third model. Notably, the lagged version of the model demonstrates a lower R-squared value

of 0.391, compared to 0.462 in the non-lagged model, indicating that including lagged variables accounts for less variance in the value of transactions.

The significant variables in the fourth model are the lagged STOXX index and Euribor. The lagged STOXX index has a coefficient of 1.922 at the 5% significance level, suggesting that stock market performance has a delayed effect on the value of M&A transactions, highlighting the influence of market conditions on M&A activities. Additionally, Euribor is significant at the 10% level with a coefficient of 7.605, indicating that interest rates also play a role in influencing the value of transactions. However, this differs from the third model, where the non-lagged Euribor rate was more significant, and the coefficient was larger. In contrast, other macroeconomic indicators, such as lagged GDP with a coefficient of 4.252, do not show a significant impact on the value of transactions, whereas the non-lagged GDP did show significance.

The adjusted R-squared also shows a decrease in the lagged model, declining to 0.334 from 0.412, reinforcing the observation that the lagged variables provide a less comprehensive explanation for the variability in transaction values. This reduced explanatory power suggests that immediate economic conditions might have a more direct influence on transaction values compared to their lagged counterparts.

Table 8 Results – Value of transactions

The results illustrate the relationship between the value of M&A transactions and economic indicators, including GDP, Euribor, STOXX, Unemployment, Exchange rate, and the impact of Covid. The analysis is conducted using two models: one considering current values of the indicators and the other using lagged values (1-month lag). Each model includes an intercept term. The results are presented with coefficients and their respective t-values in parentheses. Statistical significance is indicated by *, **, and *** representing p-values of 0.1, 0.05, and 0.01, respectively. The bottom panel of the table shows the model p-value, residual standard error, R-squared, and adjusted R-squared values.

	(3)		(4)
GDP	6,725** (2,165)	GDP_lag	4,252 (1,152)
Euribor	11,309*** (3,119)	Euribor_lag	7,605* (1,787)
STOXX	1,669*** (2,805)	STOXX_lag	1,922** (2,630)
Unemployment	23,092* (1,817)	Unemployment_lag	17,687 (1,118)
Exchange rate	0,749 (0,351)	Exchange rate_lag	1,991 (0,960)
Covid	-0,059 (-0,658)	Covid	-0,121 (-1,151)
Intercept	-94,863** (-2,150)	Intercept	-61,664 (-1,171)
Model, p-value	0,000	Model, p-value	0,000
Residual Std. Error	0,353	Residual Std. Error	0,377
R-squared	0,462	R-squared	0,391
Adjusted R-squared	0,412	Adjusted R-squared	0,334

7.3 Macroeconomic indicators' effect on the ratio of intra-European transactions

7.3.1 Normal independent variables

In the fifth regression model, examining the ratio of European target companies in M&A transactions as a dependent variable, the intercept is significantly positive at 4.156, indicating that the base ratio of European targets to non-European targets (intra-European transactions) is high when the macroeconomic indicators are at their average levels.

The model highlights a negative and significant association with GDP, presenting a coefficient of -0.241 at the 10% significance level. This finding suggests that a 1% increase in GDP is related to a 0.24% decrease in intra-European transactions, indicating that as the economy strengthens, there may be a relatively lower selection of European companies as targets in M&A. Another significant indicator is the Euribor rate, which has a negative coefficient of -0.755 at the 1% significance level, showing an inverse relationship between interest rates and the proportion of European targets. This could indicate that higher borrowing costs might

discourage acquisitions within Europe. Contrary to the Euribor's influence, the STOXX index shows no significant effect on the ratio of intra-European transactions with a coefficient close to zero (-0.059). Similarly, the unemployment rate and exchange rate do not show any significant impact on the dependent variable, with coefficients of 0.126 and 0.071, respectively.

From a model diagnostic perspective, the model is statistically significant, with a p-value of 0.037. The R-squared value at 0.181 indicates that the model explains 18.1% of the variability in the ratio of intra-European transactions, while the adjusted R-squared of 0.106 provides a lower estimate after adjusting for the number of predictors. These figures suggest that while the model captures a modest proportion of the variance, other unaccounted factors likely influence the choice of European companies as M&A targets.

7.3.2 Lagged independent variables

The sixth regression model, featuring lagged macroeconomic indicators to explain the ratio of European target companies in M&A deals, presents some changes from the previous model. The R-squared value is 0.208, demonstrating a modest and slightly better fit than the previous non-lagged model. The adjusted R-squared of 0.134 is also slightly improved from the 0.106 in the fifth model, suggesting that the lagged indicators offer a marginally better explanation of the variance when adjusting for the number of variables.

The regression results show a significant negative coefficient for the GDP at -0.320, highlighting an inverse relationship at the 5% significance level, which is more significant compared to the previous model. This suggests that past economic output also has a negative effect on M&A deals where the target is a European company. Euribor maintains its negative influence with a coefficient of -0.797, also significant at the 1% level. This indicates that higher past interest rates may discourage the selection of European targets. In contrast to these significant findings, the STOXX index shows an insignificant coefficient of -0.041, indicating that the past stock market performance does not significantly affect the choice of European targets. Unemployment and the exchange rate also have no significant impact, with coefficients of -0.521 and 0.057 respectively. These coefficients are in line with the corresponding non-lagged coefficients in the previous model and highlight their low impact on the dependent variable.

In comparing the fifth and sixth models, while the general pattern of significance across indicators remains consistent, the sixth model's use of lagged variables captures the delayed effects of economic conditions, particularly the negative impact of GDP and Euribor, on the

number of intra-European M&A transactions. In these regressions, the lagged version presents higher explanatory power as a model and the significance of the independent variables are somewhat higher.

Table 9 Results – Ratio of European targets (intra-European transactions divided by all transactions)

The results illustrate the relationship between the ratio of intra-European and all M&A transactions, and economic indicators, including GDP, Euribor, STOXX, Unemployment, Exchange rate, and the impact of Covid. The analysis is conducted using two models: one considering current values of the indicators and the other using lagged values (1-month lag). Each model includes an intercept term. The results are presented with coefficients and their respective t-values in parentheses. Statistical significance is indicated by *, **, and *** representing p-values of 0.1, 0.05, and 0.01, respectively. The bottom panel of the table shows the model p-value, residual standard error, R-squared, and adjusted R-squared values.

	(5)		(6)
GDP	-0,241* (-1,852)	GDP_lag	-0,320** (-2,575)
Euribor	-0,755*** (-2,690)	Euribor_lag	-0,797*** (-2,978)
STOXX	-0,059 (-1,072)	STOXX_lag	-0,041 (-0,747)
Unemployment	0,126 (0,163)	Unemployment_lag	-0,521 (-0,684)
Exchange rate	0,071 (0,383)	Exchange rate_lag	0,057 (0,314)
Covid	-0,012 (-1,432)	Covid	-0,015* (-1,737)
Intercept	4,156** (2,371)	Intercept	5,252*** (3,129)
Model, p-value	0,037	Model, p-value	0,018
Residual Std. Error	0,026	Residual Std. Error	0,026
R-squared	0,181	R-squared	0,208
Adjusted R-squared	0,106	Adjusted R-squared	0,134

7.4 Effects of Covid-19

Reflecting on the impact of the Covid-19 pandemic across all six regression models, the results indicate a varying degree of influence on different aspects of M&A activity, based on the specific dependent variable considered in each model. Although the significance varies, the results are the same, showing that Covid-19 impacts negatively to all of the dependent variables.

In the first set of regressions, where the number of M&A transactions is the dependent variable, Covid-19 shows a clear negative impact. Specifically, in the first regression, the Covid-19 variable has a coefficient of -0.164, significant at 1% level, suggesting that the pandemic led to a reduction in the number of transactions. In the second regression, which includes lagged

variables, Covid-19's coefficient is slightly more negative at -0.187, maintaining significance at the 1% level, indicating a continued adverse effect when considering the lagged impact.

Moving to the second set of regressions, which examine the value of M&A transactions, Covid-19's influence is similarly negative but less significant. In the third regression, the pandemic's coefficient is not statistically significant, suggesting its impact is not observable on the value of transactions. Also, in the fourth regression with lagged variables, the Covid-19 variable falls short of being significant in the accepted significance levels with a coefficient of -0.121.

In the final set of regressions focusing on the ratio of intra-European transactions to total transactions, Covid-19 shows a slight negative impact. The fifth regression presents a coefficient of -0.012 for the pandemic, which is not significant. The sixth regression, which considers lagged indicators, shows a similar pattern with Covid-19's coefficient at -0.015, now significant at the 10% level. This indicates a slight decrease in the ratio of European targets due to Covid-19 and suggests that the pandemic's effect persists even when considering past data.

Collectively, these results demonstrate that the Covid-19 pandemic has had a generally negative impact on M&A activities, affecting both the number and value of transactions and the selection of European targets. The degree of this impact varies depending on what is observed, with some models showing a more significant influence than others. The introduction of lagged variables may underline the continuing impact of the pandemic and highlight its role in shaping economic and business strategies in the M&A sector.

Table 10 Results – The impact of Covid-19

The results demonstrate the impact of Covid-19 on the number of transactions, value of transactions, and the European target. The analysis is carried out across six models: (1) and (2) focus on the number of transactions, (3) and (4) on the value of transactions, and (5) and (6) on the European target. Each column presents the coefficient of the Covid-19 variable and its corresponding t-value in parentheses. Statistical significance is denoted by *, **, and *** representing p-values of 0.1, 0.05, and 0.01, respectively. The results indicate the effect of Covid-19 on these different metrics of M&A activity.

	Number of transactions		Value of transactions		European target	
	(1)	(2)	(3)	(4)	(5)	(6)
Covid	-0,164*** (-2,769)	-0,187*** (-3,234)	-0,059 (-0,658)	-0,121 (-1,151)	-0,012 (-1,432)	-0,015* (-1,737)

7.5 Summary of the results

Table 11 Results – Summary

The combined table presents the results of the six multiple regression models analyzing the relationship between economic indicators (GDP, Euribor, STOXX, Unemployment and Exchange rate, with their respective 1-month lagged values) and the impact of Covid-19 on the number of transactions, value of transactions, and on intra-European transactions. Coefficients are presented with their t-values in parentheses, and statistical significance is denoted by *, **, and *** representing p-values of 0.1, 0.05, and 0.01, respectively.

	Number of transactions		Value of transactions		European target	
	(1)	(2)	(3)	(4)	(5)	(6)
GDP	1,990** (2,248)		6,725** (2,165)		-0,241* (-1,852)	
Euribor	-0,560 (-0,293)		11,309*** (3,119)		-0,755*** (-2,690)	
STOXX	0,774** (2,062)		1,669*** (2,805)		-0,059 (-1,072)	
Unemployment	19,194*** (3,646)		23,092* (1,817)		0,126 (-0,163)	
Exchange rate	2,111* (1,671)		0,749 (0,351)		0,071 (0,383)	
GDP_lag		0,834 (0,977)		4,252 (1,152)		-0,320** (-2,575)
Euribor_lag		0,137 (0,075)		7,605* (1,787)		-0,797*** (-2,978)
STOXX_lag		1,220*** (3,263)		1,922** (2,630)		-0,041 (-0,747)
Unemployment_lag		18,242*** (3,487)		17,687 (1,118)		-0,521 (-0,684)
Exchange rate_lag		1,401 (1,123)		1,991 (0,960)		0,057 (0,314)
Covid	-0,164*** (-2,769)	-0,187*** (-3,234)	-0,059 (-0,658)	-0,121 (-1,151)	-0,012 (-1,432)	-0,015* (-1,737)
Intercept	-29,857** (-2,501)	-14,994 (-1,301)	-94,863** (-2,150)	-61,664 (-1,171)	4,156** (2,371)	5,252*** (3,129)
Observations	72	71	72	71	72	71
Model, p-value	0,000	0,000	0,000	0,000	0,037	0,018
Residual Std. Error	0,179	0,177	0,353	0,377	0,026	0,026
R-squared	0,538	0,557	0,462	0,391	0,181	0,208
Adjusted R-squared	0,495	0,515	0,412	0,334	0,106	0,134
BP-test, p-value	0,352	0,240	0,003	0,000	0,369	0,262
JB-test, p-value	0,766	0,805	0,000	0,000	0,771	0,937
DW-test, p-value	0,165	0,103	0,922	0,521	0,928	0,677

Significance levels: *** p<1%, ** p<5%, * p<10%

To summarize the results from the regressions, GDP is generally a positive driver for the number of M&A transactions, while it has a negative impact on the ratio of European targets. This suggests that economic growth encourages M&A activity but may lead to more outward rather than inward European transactions. In contrast, the Euribor rate's significance is

mixed. Its significance is generally low in explaining the M&A activity, except for explaining the value of M&A. Explaining the intra-European M&A, it has a high inverse effect on the dependent variable.

The stock market performance shown by the STOXX index is positively correlated with both the number and value of transactions, with high statistical significance. This correlation may indicate that stock market conditions influence M&A activity, which may be due to factors such as investor confidence and the availability of capital for M&A transactions. Unemployment generally shows a significant positive correlation with the number and value of M&A transactions, which could point to strategic consolidations during economic downturns when unemployment is high. The exchange rate shows a significant effect only in model 1 model, suggesting currency fluctuations might not be a major consideration in M&A decision-making.

Finally, Covid-19 has a consistently negative impact on both the number and value of transactions and the ratio of European targets, reflecting the pandemic's effects on economic and corporate activities. The overall model diagnostics indicate good model fit and significance, yet while these macroeconomic factors are important, other variables not included in the models may also play a substantial role in explaining the M&A activity. An alternative result summary Table 12 is presented below, which illustrates the direction of the relationship, if significant. A plus sign indicates that change in the independent variable affects the dependent variable positively while minus indicates an inverse effect. NS stands for “not significant”.

Table 12 Results – Summary of the relationship direction

	Number of transactions		Value of transactions		European target	
	(1) Non-lagged	(2) Lagged	(3) Non-lagged	(4) Lagged	(5) Non-lagged	(6) Lagged
GDP	+	NS	+	NS	-	-
Euribor	NS	NS	+	+	-	-
STOXX	+	+	+	+	NS	NS
Unemployment	+	+	+	NS	NS	NS
Exchange rate	+	NS	NS	NS	NS	NS
Covid	-	-	NS	NS	NS	-
Intercept	-	NS	-	NS	+	+

Note: Columns represent the regressions 1-6, +/- only if the explanatory variable significant

8 DISCUSSION

In this chapter, the results from the previous chapter will be discussed in more detail. These findings will be connected to the theoretical framework and existing literature discussed earlier in the study.

8.1 The macroeconomic indicators' effect on M&A

8.1.1 *Gross domestic product*

When discussing the relationship between GDP growth and M&A activity in line with the first hypothesis (H1), the regression results support the hypothesis that GDP growth has a positive effect on the number of M&A transactions, with a significant coefficient 1.990 in model 1. This aligns with Stoian & Filippaios (2008) findings that the positive relationship is supported by economies of scale and increased market demand as the economy grows, suggesting that firms leverage improved financial environment and they are also motivated to diversify and manage risk by M&A (Globerman & Shapiro, 1999; Uddin & Boateng, 2011). This is reflected in number of transactions, indicating that a healthy economy encourages firms to pursue growth through M&A.

The relationship between GDP and the value of transactions also presents a positive coefficient 6.725, suggesting that GDP growth does indeed increase the value of transactions. This supports the hypothesis that economic growth not only increases the number of M&A deals but also the scale of these transactions, which could be due to firms leveraging economies of scale and exploiting new opportunities in an expanding market.

However, when examining the normal and lagged effects of GDP on the ratio of European target companies, the results from models 5 and 6 with coefficients of -0.241 and -0.320 respectively, indicate a negative relationship. This suggests that while firms may be engaging in more transactions and at higher values, they are not necessarily targeting European firms more as the economy grows. This could mean that economic growth may encourage international diversification, as companies seek opportunities outside of their domestic markets.

8.1.2 *Interest rates*

In evaluating the impact of interest rates on M&A activity, the second hypothesis (H2) states a negative relationship, suggesting that higher interest rates would likely discourage M&A transactions due to increased financing costs. The 3-month Euribor rate shows significant effects only in models 3 and 4, where unexpectedly positive relationships with the value of

M&A transactions are observed at the 1% and 10% significance levels, respectively. This result contrasts with the hypothesized negative relationship and challenges the view that higher interest rates discourage M&A activity. It suggests instead that an increase in the Euribor rate is associated with a rise in the value of transactions.

The results challenge the arguments put forward by Moeller and Schlingemann (2005) that higher interest rates may discourage M&A due to higher financing costs, as well as Forssbaeck and Oxelheim (2008), who argue that cost and access to capital are major factors for the ability to undertake cross-border M&A. Instead, the results of this study suggest that the role of interest rates on M&A transactions should be reconsidered. It may be that higher interest rates not only reflect financing costs, but also indicate expectations of future economic growth or inflationary pressures, which could encourage firms to lock in current valuations through M&A.

Since the results for the Euribor rate in the models explaining the number of M&A transactions do not show significant relationship, it suggests that the overall economic environment, company-specific strategies, or other macroeconomic indicators are more important. This lack of significant effects challenges the findings of previous studies and supports the argument that factors beyond interest rates and the cost of capital might have a greater impact on M&A activity.

In models 5 and 6, the Euribor rate showed a significant negative effect on the ratio of intra-European transactions. This outcome suggests that as interest rates rise, there is a relative decline in the M&A activity targeting European companies compared to those targeting companies outside Europe. This result can be rationalized by the findings of other studies, as higher interest rates raise the cost of capital, reduce the investment attractiveness of European targets, and increase risk aversion, which discourages M&A activity within Europe by making acquisitions more expensive and less appealing.

8.1.3 The stock market

In addressing the third hypothesis (H3), which examines the relationship between stock market performance, as indicated by the STOXX600 index, and M&A activity, the empirical findings support the hypothesis that a better performing stock market in Europe has a significant positive relationship with M&A activity, whether considering normal or lagged STOXX600 values.

This finding is in line with works of Weston (1953) and Nelson (1959), who found a positive correlation between stock prices and M&A activity. Specifically, this positive relationship with the number and value of M&A suggests that a booming stock market drive M&A activity, as suggested by Shleifer and Vishny (2003) and Baker et al. (2009). This is consistent with their overvaluation hypothesis, where managers utilize high-valued stock as currency to acquire other companies, hedging against future stock price corrections.

Furthermore, the lack of a significant relationship between the STOXX index and the ratio of European targets indicates that stock prices may not significantly impact the geographical direction of M&A. This could suggest that other factors, such as strategic considerations or market-specific other factors, are more crucial in determining whether European companies engage in M&A activities in Europe or outside of Europe.

8.1.4 The unemployment rates

Exploring the relationship between unemployment rates and M&A activity presents a less-studied perspective within the context of M&A. The fourth hypothesis (H4) suggests a negative relationship, meaning that that rising unemployment rates, reflecting economic downturns, may reduce M&A activity due to decreased consumer spending, lower corporate revenues, and increased financial uncertainty.

Contrary to this hypothesis, the regression models 1, 2 and 3, reveal an interesting finding, showing that unemployment rates are positively correlated with both the number and value of M&A transactions, with significant coefficients of 19.194, 18.242 and 23.092, respectively. These results suggest that rather than hedging risk, firms might be leveraging M&A as a strategic tool during economic downturns. This could be to possibly gain market share, access new technologies, or restructure operations for improved efficiency. This pattern does not support the expected negative correlation and instead may imply that firms are using M&A as a sort of counter-cyclical strategy. Furthermore, these findings challenge the theoretical argument that high unemployment would discourage M&A due to reduced purchasing power and profits. While significant results regarding these findings arise in this study, unemployment rates should be further studied in order to draw reliable conclusions.

The lack of a significant relationship between unemployment and the ratio of European targets in models 5 and 6 suggests that while unemployment may affect the overall M&A market, it does not specifically influence whether firms target European or non-European companies.

8.1.5 The exchange rates

Examining the influence of exchange rates on M&A activity brings us to the fifth and sixth hypotheses, which are based on the overvaluation hypothesis and the dynamics of currency strength in cross-border acquisitions. The hypotheses propose that a stronger domestic currency, as indicated by a strengthening USD/EUR exchange rate, should have a positive effect on overall M&A activity. This is due to more favorable acquisition costs and overall economic conditions (H5), and a negative effect on intra-European M&A activity as European assets become relatively more expensive for foreign assets (H6).

The empirical evidence from the regression models, however, presents interesting findings where exchange rate fluctuations show significant effects only on the number of M&A transactions when non-lagged, and insignificance on value of transactions and the ratio of European targets. Across all other models than the model 1, the exchange rate coefficients are not statistically significant, which challenges the hypotheses based on studies by Boateng et al. (2014) and Erel et al. (2012). Also, studies such as Froot and Stein (1991) and Goergen and Renneboog (2004) suggest that companies from countries with stronger currencies tend to be more active acquirers in international markets. Still, the results of this study suggest that, contrary to expectations, the exchange rate may not be a primary determinant in M&A decision-making or that its impact is potentially diminished by other factors.

Furthermore, the non-significant impact of exchange rates on the preference for European targets in models 5 and 6 suggests that factors other than currency strength, for example market potential, synergy gains, or geopolitical considerations, could be driving the geographical focus of M&A activity. This observation may indicate a change from the previous findings by Froot and Stein (1991) and Goergen and Renneboog (2004), where currency strength was linked to international acquisition behavior.

8.1.6 Non-lagged versus lagged indicators

The seventh hypothesis (H7) addresses expectation that lagged macroeconomic indicators might be more significant predictors of M&A activity than non-lagged indicators. This hypothesis is based on the rationale of M&A decision-making, which is not based on immediate economic conditions but on a thorough strategic assessment and planning that includes a temporal lag.

Observing the regression results, mixed support for this hypothesis is found. Lagged variables such as STOXX and Unemployment rate do show significant explanatory power in relation to

the number and value of M&A transactions in this study and the R-squared value for model 2, which represents the lagged effects of the indicators to the number of M&A, is higher than for model 1 using non-lagged indicators. Also, model 6, which specifically focuses on the ratio of European targets, indicates that lagged GDP and Euribor provide significant explanatory power, that is higher for the lagged model. These results support the formulated hypothesis and the findings of Boateng et al. (2014), Uddin and Boateng (2011) and Bhagwat et al. (2016), which highlight a delayed response to economic changes. However, the hypothesis is not supported across all the economic indicators. For instance, lagged GDP and Euribor do not show explanatory significance as high in their lagged forms when observing the number and value of transactions. This may indicate that certain economic influences, such as overall economic growth, might have both immediate and delayed impacts on M&A activity, or that their influence may be reflected differently depending on a specific M&A strategy or market conditions.

8.1.7 Positive domestic changes yield more domestic activity

In the formulation of the eighth hypothesis (H8), the focus is on the influence of positive changes in macroeconomic factors on M&A transactions in Europe (referred to as domestic). This is based on the assumption that a strong macroeconomic environment improves the financial performance and prospects of companies, which contributes to M&A activity by improving the attractiveness of companies and reducing investment risks.

The results from the regression reveal opposite findings. While GDP, interest rates, stock market performance and unemployment rates are mainly positively related to the number and value of M&A transactions, the indicators are negatively associated with the ratio of European target companies. This suggests that while a strong economy may lead to an overall increase in M&A activities, it does not necessarily make European companies more attractive targets, potentially due to an expansion of opportunities that includes more attractive investments outside of Europe.

However, the interest rates show results that support the formulated hypothesis, as lower values can be seen as positive for M&A activity. A negative coefficient for Euribor rates in relation to the ratio of European targets suggests that as interest rates decrease, the proportion of M&A transactions targeting European firms increases, relative to non-European targets. This indicates that lower interest rates may encourage more European M&A activities. Additionally, when considering lagged unemployment rates, lower levels, which can be seen as a reflection of economic health, are negatively correlated with M&A activity in Europe, although this finding is not statistically significant. This would mean that lowering

unemployment rates would increase the M&A activity in Europe, but only if stronger statistical correlation could be found.

8.2 Effects of Covid

The ninth hypothesis (H9) states that the Covid-19 pandemic has impacted European as well as overall M&A activity, because external shocks often force companies to reassess their strategic objectives and could influence their engagement in M&A.

The regression results confirm this hypothesis, indicating a negative effect of the Covid-19 pandemic across various dimensions of M&A activity. Notably, the pandemic's impact is seen in the negative coefficients associated with the Covid-19 dummy variable in all regression models for both the number and value of transactions. This aligns with the observations made by Andrade et al. (2001) and Mulherin and Boone (2000), who noted that merger activities are influenced by economic and regulatory shocks. The pandemic, as a major external shock, appears to have affected M&A strategies, likely due to higher uncertainty, regulatory changes, and shifts in consumer behavior.

Furthermore, the results align with Choi and Jeon (2011), who state that economic growth cycles impact M&A activity, indicating that an economic downturn caused by the pandemic could reduce such activities. Cerrato et al. (2016) also support this view by showing that firms often focus on their core business and domestic markets during economic downturns, leading to a decrease in M&A activities, especially international ones.

9 CONCLUSION

This paper has systematically examined the influence of macroeconomic indicators on M&A activity, revealing interdependencies and varied impacts. The analysis demonstrates that GDP growth strongly stimulates M&A, both in terms of transaction frequency and value, confirming that economic health, reflected by growth, is a critical driver of corporate expansion through M&A. This positive relationship suggests that firms leverage favorable economic conditions to explore growth opportunities and manage risks through diversification.

Contrary to initial hypotheses, the study finds that higher interest rates, represented by the 3-month Euribor, correlate positively with the value of M&A transactions. This finding suggests that higher interest rates may coincide with periods of economic optimism or strategic acquisitions, challenging the traditional view that they solely discourage them by increasing financing costs. This indicates a more complex role of interest rates, possibly reflecting broader economic conditions or strategic imperatives rather than direct cost implications alone.

The analysis also highlights the significant role of stock market performance in driving M&A activity, aligning with previous literature suggesting that a robust stock market provides the capital and confidence necessary for companies to engage in acquisitions. Interestingly, higher unemployment rates are found to correlate with increased M&A activity, suggesting that firms might strategically utilize acquisitions during economic downturns for restructuring or gaining market share, rather than withdrawing from these activities.

When observing the ratio of intra-European acquisitions explained by European macroeconomic indicators, findings show that while favorable conditions, such as lower interest rates, encourage more European acquisitions, broader economic growth leads companies to seek opportunities outside of Europe. This suggests that while economic strength supports domestic acquisitions under specific conditions, it also motivates firms to pursue global expansion.

Finally, the Covid-19 pandemic has had a strong negative impact on M&A across various metrics during the observed period of 2017-2022, reflecting the disruptions and uncertainties introduced by the global health crises. Overall, this study has shown that macroeconomic indicators have a significant impact on the M&A activity enriching the understanding of how different macroeconomic environments influence corporate M&A strategies, providing insights that challenge traditional theories and highlighting the complexity of financial decision-making in the globalized market.

9.1 Limitations of the study

There are limitations in the study, which are outlined in this section. First, there are data limitations, as the analysis is based on transactions reported by Bloomberg. Not all acquisitions in the target area may necessarily appear in the sample if they are not registered in Bloomberg, which may lead to an incomplete market representation. Secondly, Eurostat's macroeconomic indicators may be affected by revisions or methodological changes in data reporting, even though the data is collected with the same specifications for the different indicators. Converting quarterly GDP figures into monthly estimates leads to an assumption of a flat economic activity distribution, which may not reflect the actual monthly fluctuations.

The geographical focus on Europe limits the applicability of the study to the wider global context, where external economic forces can affect regional business activity. Furthermore, the time span of the study, limited to the period before and during the Covid-19 period, may not capture the full extent of the cyclical and dynamic nature of M&A activity. Finally, the potential impact of regulatory changes and the fact that nearly the entire studied period was a zero-interest rate period, excluding the "normal" period, must be taken into account.

Variable selection creates a limitation for the study, as there might be certain macroeconomic variables that significantly explain variations in M&A activity but are not included in this study.

9.2 Suggestions for future research

For further research on the topic of macroeconomic influences on M&A activity, a variety of approaches could be explored to deepen understanding and broaden the context of current findings. An analysis across different geographical regions could reveal how macroeconomic indicators influence M&A activity uniquely in diverse economic settings focusing on the Nordic region, for example. Additionally, investigating how different industries respond to economic changes might reveal some sector-specific dynamics in M&A behavior, given varying sensitivities to economic cycles.

Longitudinal studies, covering multiple years or even decades, that track the effects of macroeconomic changes on M&A over extended periods, could be utilized. This might be valuable, providing insights into the lagged and long-term impacts of economic fluctuations on corporate strategies.

The influence of political stability and governmental economic policies on M&A could offer another layer of understanding, particularly how political transitions and economic reforms might create or constrain opportunities for acquisitions. A detailed analysis of financing methods for M&A in response to macroeconomic conditions could also enhance the understanding of how firms finance acquisitions differently in varying economic environments.

Lastly, exploring a broader range of variables that describes the macroeconomic environment could further deepen the understanding of the determinants of M&A activity. Including variables such as political stability, technological readiness, or industry-specific factors could allow an even more comprehensive analysis of the drivers behind M&A decisions.

9.3 The use of artificial intelligence in the text

In the preparation and finalization of this thesis, two advanced language models were utilized: ChatGPT by OpenAI (specifically the GPT-4 model, 2024) and Gemini (2024) by Google (previously known as Bard). These tools were used to refine the language and structure of the thesis. I wrote each section independently to ensure the content reflected my understanding and original insights. After completing each draft, I used ChatGPT or Gemini to review and enhance the text and analyzed myself which one provided better suggestions. The interaction involved submitting the text sections for feedback, with A.I. to provide suggestions to improve clarity, coherence, academic style and overall quality. The prompts used were designed to help align my writing style throughout the thesis and to ensure the enhancements did not alter the original content or introduce new information. The used prompts were the following with slight modifications depending on the exact situation:

“Is the following text grammatically correct? [Input]”

“Analyze the coherence and academic style in the following text. [Input]”

“Suggest a new version of the provided text utilizing the improvement ideas. Focus solely on the given text and do not add information. Avoid using intensifying words but maintain an academic style.”

The use of ChatGPT and Gemini was strictly limited to language enhancement and structural editing. All ideas, analysis, and conclusions presented in this thesis are entirely my own, based on my research and understanding of the topic. These tools were essential in polishing the final manuscript while preserving the integrity and originality of my work.

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APPENDIX 1

	VIF values	
	Non-lagged	Lagged
GDP	5,868	5,425
Euribor	5,764	5,259
STOXX	2,802	2,837
Unemployment	2,673	2,630
Exchange rate	2,704	2,646
Covid	1,952	1,892